PHYSICS

1) A particle starts from rest with constant acceleration. The ratio of space–average velocity to the time average velocity is :

(1) $\frac{1}{2}$

(2) $\frac{3}{4}$

(3) $\frac{4}{3}$

(4) $\frac{3}{2}$

2) Let $\vec{v}$ and $\vec{a}$ denote the velocity and acceleration respectively of a body in one-dimensional motion

(1) $|\vec{v}|$ must decrease when $\vec{a} < 0$

(2) speed must increase when $\vec{a} > 0$

(3) speed will increase when both $\vec{v} < 0$ and $\vec{a} > 0$

(4) speed will decrease when $\vec{v} < 0$ and $\vec{a} > 0$

3) A person first goes $10\sqrt{3}$ m in East direction. Then after he goes 10 m in north direction and finally he goes $20\sqrt{3}$ m 60° south of west. Net displacement of person will be :-

(1) 20 m, in north

(2) 10 m, in north

(3) 20 m, in south

(4) 10 m, in south

4) A body starts from rest with uniform acceleration. What is the ratio of distance travelled during the 4th and 3rd sec.

(1) $\frac{7}{5}$

(2) $\frac{5}{7}$

(3) $\frac{7}{3}$

(4) $\frac{3}{7}$

5) The velocity v of a moving particle varies with displacement as $x = \sqrt{v+1}$. The acceleration of the particle at x = 5 unit will be :-

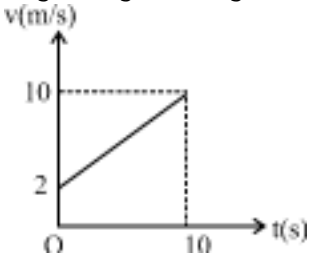
(1) $\sqrt{6}$ unit

(2) 24 unit

(3) 240 unit

(4) 25 unit

6) The velocity time graph of a car moving along a straight road is shown in figure. The average velocity of car in first 10 second is :-



(1) 2 m/s

(2) 4 m/s

(3) 6 m/s

(4) 8 m/s

7) A man wishes to swim across a river 0.5 km wide. If he can swim at the rate of 2 km/h in still water and the river flows at the rate of 1 km/h. The angle made by the directon (w.r.t. the flow of the river) along which he should swim so as to reach a point exactly opposite his starting point, should be :

(1) 60°

(2) 120°

(3) 145°

(4) 90°

8) A particle starts from rests and travels a distance S with uniform acceleration, then it travels a distance 2S with uniform speed, finally it travels a distance 3S with uniform retardation and comes to rest. If the complete motion of the particle is a straight line then the ratio of its average velocity to maximum velocity is :

(1) $\frac{6}{7}$

(2) $\frac{4}{5}$

(3) $\frac{3}{5}$

(4) $\frac{2}{5}$

9)

A particle moves along x-axis as

$x = 4(t-2) + a(t-2)^2$

Which of the following is true ?

(1) The initial velocity of particle is 4

(2) The acceleration of particle is 2a

(3) The particle is at origin at t = 0

(4) None of these

10) A particle has an initial velocity of $3\hat{i} + 4\hat{j}$ and an acceleration of $0.4\hat{i} + 0.3\hat{j}$. It's speed after 10 sec is :-

(1) 7 units

(2) 8.5 units

(3) 10 units

(4) $7\sqrt{2}$ units

11) A body is projected vertically upwards with a velocity of 10 ms^{-1}. It reaches the maximum vertical height h in time t. In time t/2, the height covered is :-

(1) $\frac{h}{2}$

(2) $\frac{2h}{5}$

(3) $\frac{3h}{4}$

(4) $\frac{5h}{8}$

12) A body is dropped from height of 100 m. At what height from ground the velocity of the body will be equal to half of velocity when it hits the ground :

(1) 45 m

(2) 55 m

(3) 65 m

(4) 75 m

13) The co-ordinates of a moving particle at any time t are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time t is given by :-

(1) $\sqrt{\alpha^2 + \beta^2}$

(2) $t^2\sqrt{\alpha^2 + \beta^2}$

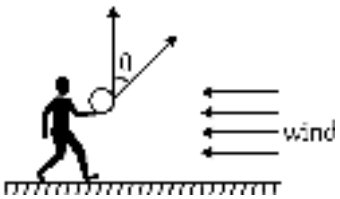
(3) $3t\sqrt{\alpha^2 + \beta^2}$

(4) $3t^2\sqrt{\alpha^2 + \beta^2}$

14)

A boy throws a ball upwards with velocity v_0 = 20 m/s. The wind imparts a horizontal acceleration of 4 m/s^2 to the ball. The angle θ from vertical at which the ball must be thrown so that the ball returns

to the boy's hand is ($g = 10\ m/s^2$)



(1) $\tan^{-1}(1.2)$

(2) $\tan^{-1}(0.2)$

(3) $\tan^{-1}(2)$

(4) $\tan^{-1}(0.4)$

15) A stone is projected with speed u and angle of projection is θ. Find radius of curvature at t = 0.

(1) $\frac{u^2\cos^2\theta}{g}$

(2) $\frac{u^2}{g\sin\theta}$

(3) $\frac{u^2}{g\cos\theta}$

(4) $\frac{u^2\sin^2\theta}{g}$

16) An express train is moving with a velocity v_1. Its driver finds another train is moving on the same track in the same direction with velocity v_2. To escape collision, driver applies a retardation a on the train. the minimum time of escaping collision will be :-

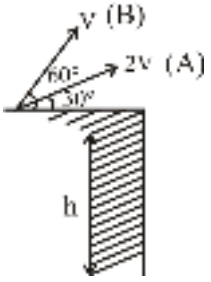
(1) $t = \frac{v_1 - v_2}{a}$

(2) $t_1 = \frac{v_1^2 - v_2^2}{2}$

(3) None

(4) Both

17) Two particles A & B are projected from a building. A is projected with speed 2V making an angle 30° with horizontal & B with speed V making an angle 60° with horizontal. Which particle will hit the ground earlier ?



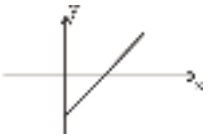
(1) Particle A

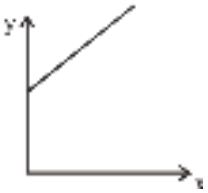
(2) Particle B

(3) Particle A & B will hit at same time.

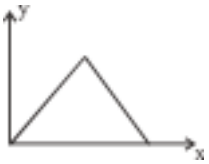
(4) None of these

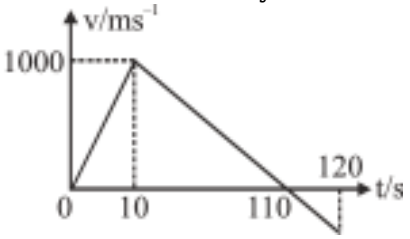
18) If y = |x–1|, the y-x graph is :-

(1) 

(2) 

(3) 

(4) 

19) The graph shows the variation of velocity of a rocket with time. Then, the maximum height attained by the rocket is 

(1) 1.1 km
(2) 5 km
(3) 55 km
(4) None of these

20) A projectile is given initial velocity of $\hat{i} + 2\hat{j}$. The equation of the path is : (g = 10 m/s^2)

(1) $y = 2x - 5x^2$
(2) $y = x - 5x^2$
(3) $4y = 2x - 5x^2$
(4) $y = 2x - 25x^2$

21) $\vec{A}$ and $\vec{B}$ are always :-

(1) Parallel vectors
(2) Equal vectors
(3) Collinear vectors
(4) Coplanar vectors

22) Find value of $\left(2\vec{A}+3\vec{B}\right).\left(3\vec{A}-2\vec{B}\right)$

(1) zero

(2) $6AB\cos\theta + 9A^2 - 9B^2$

(3) $6A^2 + 5AB\cos\theta - 6B^2$

(4) $5AB\cos\theta$

23) Which of the following is not a vector :-

(1) $\vec{A} + \vec{B}$

(2) $\vec{A} \cdot \vec{B}$

(3) $\vec{A} - \vec{B}$

(4) $\vec{A} \times \vec{B}$

24) If $\vec{A} = 5\hat{i} - \hat{j} + 7\hat{k}$ and $\vec{B} = -2\hat{i} + 3\hat{j} - \hat{k}$
then find out a unit vector parallel to resultant vector of $\vec{A}$ and $\vec{B}$:-

(1) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$

(2) $3\hat{i} + 2\hat{j} + 6\hat{k}$

(3) $\frac{3\hat{i}+2\hat{j}+6\hat{k}}{7}$

(4) $\hat{i} + \hat{j} + \hat{k}$

25) If $\left|\hat{a}+\hat{b}\right| = 2$ then $\left|\hat{a}-\hat{b}\right|$ equals :-

(1) 0

(2) 1

(3) $\sqrt{3}$

(4) $\sqrt{2}$

26) Which of the following cannot be the vector addition of these vectors. If two forces are 10N and 15N ?

(1) 4 N

(2) 5 N

(3) 5.5 N

(4) 24.5 N

27) If $\vec{A}$ and $\vec{B}$ are two non-zero vectors such that $|\vec{A}+\vec{B}| = \frac{|\vec{A}-\vec{B}|}{2}$ and $|\vec{A}| = 2|\vec{B}|$ then the angle between $\vec{A}$ and $\vec{B}$ is :

(1) 37°

(2) 53°

(3) $\cos^{-1}(-3/4)$

(4) $\cos^{-1}(-4/3)$

28) A particle starts moving with acceleration 2 m/s^2. Distance travelled by it in 5th half second is :-

(1) 1.25 m

(2) 2.25 m

(3) 6.25 m

(4) 30.25 m

29) Magnetic flux ϕ (in weber) in a closed circuit of resistance 10 ohm varies with time t (in sec) according to the equation $\phi = 6t^2 - 5t + 1$. What is the magnitude of the induced current at t = 0.25 s ?

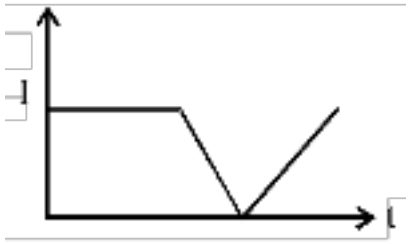
(1) 1.2 A

(2) 0.8 A

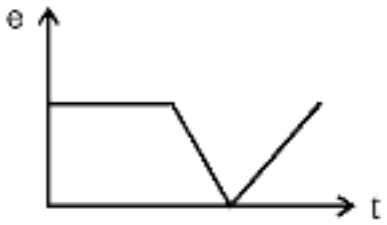
(3) 0.6 S

(4) 0.2 A

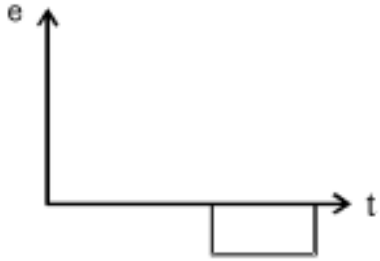
30) i–t graph for a coil is shown in this figure. which of the following graph shows the induced e.m.f. with time :-

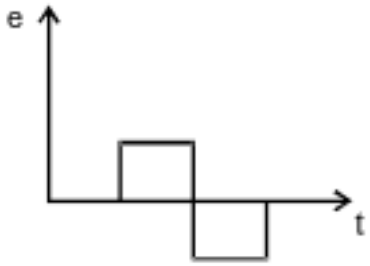


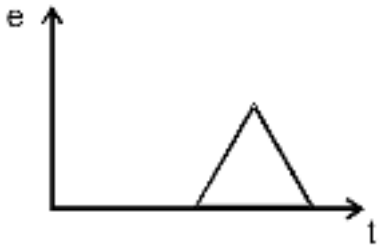
(1)

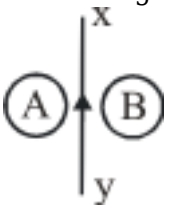


(2)



(3) 

(4) 

31) Consider the situation shown in figure. If the current I in the long straight wire xy increases at a steady rate then induced current in loop A and B will be :- 

(1) Clockwise in A and anticlockwise in B

(2) Anticlockwise in A and clockwise in B

(3) Clockwise in both A and B

(4) Anticlockwise in both A and B

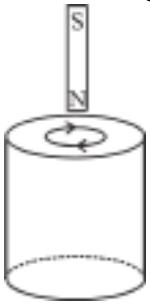
32) A solenoid of radius R and length L has a current $I = I_0 \cos\omega t$. The value of induced electric field at a distance of r outside (from axis) the solenoid, is :

(1) $\frac{\mu_0 n I_0 \omega R^2}{2r} \sin\omega t$

(2) $\frac{\mu_0 n I_0 \omega R^2}{r} \sin\omega t$

(3) $\frac{\mu_0 n I_0 R^2}{2r} \sin\omega t$

(4) Zero

33) When a magnet is released from rest along the axis of a hollow conducting cylinder situated vertically as shown in figure, then: 

(1) The direction of induced current in the cylinder is clockwise as seen from the figure

(2) The magnet moves with an acceleration more than g = 9.8 m/s^2

(3) The cylinder gets heated

(4) The speed of the magnet goes on increasing continuously if the cylinder is very long

34) Coefficient of self induction becomes $2\sqrt{2}$ times for a plane circular coil when its radius will changes, keeping its number of turns same, then area of coil becomes how much times :-

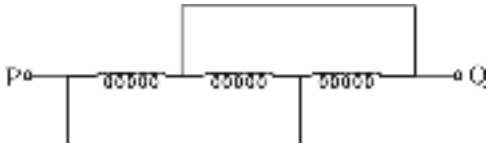
(1) 8 times

(2) 4 times

(3) 2 times

(4) 16 times

35) Pure inductors each of inductance 3 H are connected as shown. The equivalent induction of the circuit is :-



(1) 1H

(2) 2H

(3) 3H

(4) 9H

36) The self inductance of a toroid is :-
(N: no. of turns, r: radius of cross section, R_m: mean radius of toroid) :-

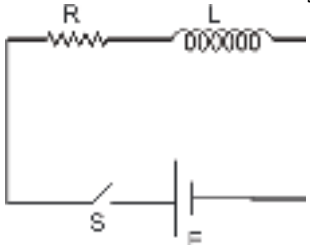
(1) $\frac{\mu_0 N^2 r^2}{2 R_m}$

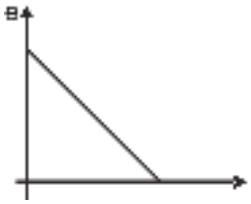
(2) $\frac{\mu_0 N^2 \pi r}{2R_m}$

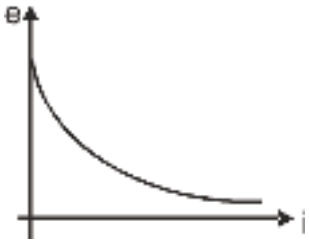
(3) $\frac{\mu_0 N^2 r}{2R_m}$

(4) $\frac{\mu_0 N^2 r\pi}{R_m}$

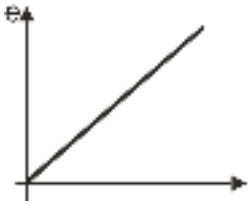
37) In an L-R circuit connected to a battery of constant e.m.f. E switch S is closed at time t = 0. If e denotes the induced e.m.f. across inductor at any time t. Then which of the following graphs shows the variation of e with t ?

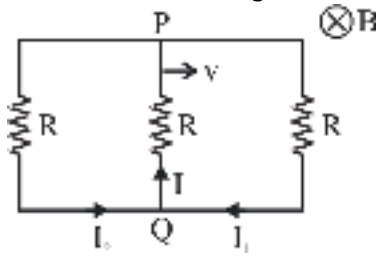


(1) 

(2) 

(3) 

(4) 

38) A rectangular loop has a sliding connector PQ of length and resistance R and is moving with a speed v as shown in the figure. The set up is placed in a uniform field going into the plane of the paper. 

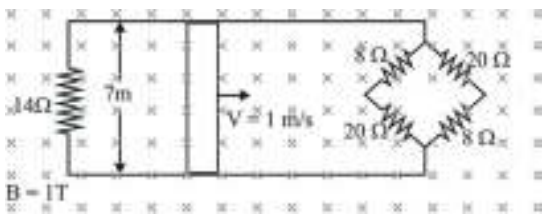
The three currents are :-

(1) $I_1 = I_2 = \frac{B\ell v}{6R};\ I = \frac{B\ell V}{3R}$

(2) $I_1 = -I_2 = \frac{B\ell v}{R};\ I = \frac{2B\ell V}{R}$

(3) $I_1 = I_2 = \frac{B\ell v}{3R};\ I = \frac{2B\ell V}{3R}$

(4) $I_1 = I_2 = I = \frac{B\ell V}{R}$

39) 

In above circuit, Resistance of moving rod is zero, then find out current through rod :-

(1) 1 A

(2) 2 A

(3) 0.5 A

(4) 0.25 A

40) In a transformer, there are 500 turns in primary coil and 5000 turns in secondary coil. A source of 80V and 50 Hz has been connected to it. Then what will be potential and frequency in the secondary coil ?

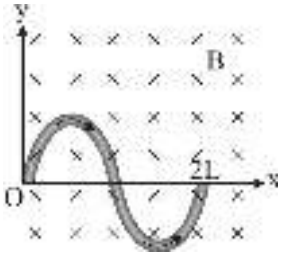
(1) 80 V and 25 Hz

(2) 80 V and 50 Hz

(3) 8 V and 50 Hz

(4) 800 V and 50 Hz

41) A wire carrying a current i is placed in a uniform magnetic field in the form of the curve



$y = a\sin\left(\frac{\pi x}{L}\right), 0 \leqslant x \leqslant 2L.$ The force acting on the wire is :-

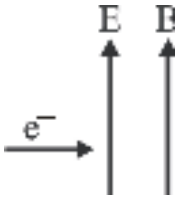
(1) $\frac{iBL}{\pi}$

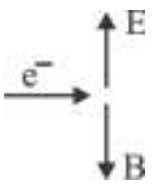
(2) $iBL\pi$

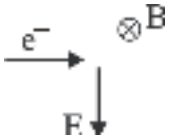
(3) $2iBL$

(4) Zero

42) An electrostatic field E and a magnetic field B act in the same region. Which one of the combination shown in the figure can cause the electrons to pass undeflected?

(1)


(2)


(3)


(4)


43) A current i is flowing in a conductor of length □. When it is bent in the from of a circular loop

then its magnetic moment will be :-

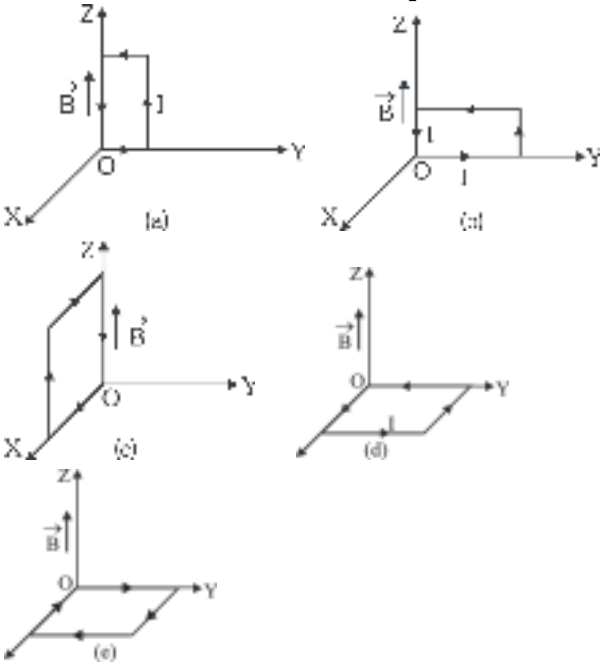
(1) $\frac{i\ell^2}{4\pi}$

(2) $\frac{i\ell}{4\pi}$

(3) $\frac{4\pi}{\ell^2 i}$

(4) $4\pi\ell^2 i$

44) A uniform magnetic field of 5000 gauss is established along the positive z-direction. A rectangular loop of side 20 cm and 5 cm carries a current of 10 A is suspended in this magnetic field. Which case/cases corresponds to stable equilibrium?



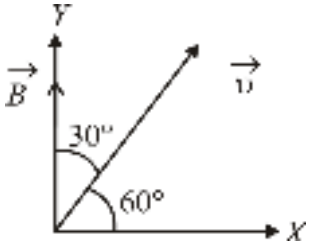
(1) a,b,c

(2) c,d,e

(3) d,e

(4) d

45) A proton of mass 1.67×10^{-27}kg and charge 1.6×10^{-19}C is projected with a speed of 2×10^6 m/s at an angle of 60° to the X-axis as shown. If a uniform magnetic field of 0.104 Tesla is applied along Y-axis, the path of proton is



(1) A circle of radius 0.2 m and time period $\pi \times 10^{-7}$s

(2) A circle of radius 0.1 m and time period $2\pi \times 10^{-7}$s

(3) A helix of radius 0.1 m and time period $2\pi \times 10^{-7}$s

(4) A helix of radius 0.2 m and time period $4\pi \times 10^{-7}$s

CHEMISTRY

1) When cells of skeletal vacuoles of a frog were placed in a series of NaCl solution of different concentration solution at 27°C, it was observed microscopically that they remained unchanged in 0.7% solution, shrank in a more concentrated and swelled in more dilute solutions, 0.7% freezes salt solution at –0.4°C. What is the osmotic pressure of the cell cytoplasm at 27°C. (K_f = 1.8 kg mol^{-1} K). Consider molarity and molality same for salt solution.

(1) 0.29 atm

(2) 2.81 atm

(3) 3.79 atm

(4) 5.55 atm

2) The osmotic pressure of 0.4% $\frac{w}{v}$ urea solution is 1.64 atm and that of 3.42% $\frac{w}{v}$ sucrose solution is 2.46 atm. When equal volume of the above two solutions are mixed, the osmotic pressure of the resulting solution is : (Temperature is not room temperature)

(1) 1.64 atm

(2) 2.46 atm

(3) 4.10 atm

(4) 2.05 atm

3) Which of the following statements is correct regarding a solution of two component A and B exhibiting negative deviation from ideal behaviour ?

(1) $\Delta_{mix}H \geq 0$ at constant T and P

(2) Intermolecular attractive forces between A-A and B-B are stronger than those between A-B

(3) $\Delta_{mix}V \geq 0$ at constant T and P

(4) Intermolecular attractive forces between A-A and B-B are weaker than those between A-B

4) **Statement I :** Higher the molal depression constant of solvent used, higher the freezing point of solution.
Statement II : Depression in freezing point does not depend on the nature of solvent.

(1) Both statement-I and statement-II are correct

(2) Statement-I is correct and statement-II is incorrect

(3) Statement-I is incorrect and statement-II is correct

(4) Both statement-I and statement-II are incorrect

5) Which of the following aqueous solution will exhibit highest boiling point ?

(1) 0.01 M Urea

(2) 0.01 M KNO_3

(3) 0.01 M Na_2SO_4

(4) 0.015 M $C_6H_{12}O_6$

6) **Assertion (A) :** The osmotic pressure of 0.1 M urea solution is less than 0.1 M NaCl solution.
Reason (R) : Osmotic pressure is not a colligative property.

(1) Both Assertion and Reason are true

(2) Both Assertion and Reason are false

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

7) The concentration of solutions A, B, C and D are respectively 0.1M glucose, 0.05 M KCl, 0.05M $MgCl_2$, 0.1 M $AlCl_3$. Which pair amongst these solutions is isotonic :-

(1) A and B

(2) B and C

(3) A and D

(4) A and C

8) The normal freezing point of nitrobenzene is 278.82 K. If a non-volatile solute is dissolved in it and a solution of 0.25 molality is prepared. The observed freezing point of the solution is 276.82 K. Evalute K_f of nitrobenzene ?

(1) 6 K $mole^{-1}$ kg

(2) 8 K $mole^{-1}$ kg

(3) 7 K $mole^{-1}$ kg

(4) 5 K $mole^{-1}$ kg

9) Two solutions containing respectively 0.60 gm of urea (mol. wt. 60) and 1.2 g of compound X in 100 gm water, freezes at the same temperature. Calculate molecular weight of X.

(1) 100

(2) 120

(3) 140

(4) 160

10) The term which is equal to relative lowering of vapour pressure (RLVP), is

(1) $\frac{P^o - P_S}{P_S}$

(2) $P°–P_S$

(3) X_{solute}

(4) $X_{solvent}$

11) The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order :

(1) B > A > C

(2) B > C > A

(3) A > C > B

(4) A > B > C

12) 36.5% by mass HCl solution has density 1.2 g mL^{-1}, molarity of solution is

(1) 3 M

(2) 8 M

(3) 10 M

(4) 12 M

13) Check the following statements :
(i) A catalyst does not alter the equilibrium state.
(ii) For the equilibrium, $H_2 + I_2 \rightleftharpoons 2HI$, the value of equilibrium constant increases with increase in concentration of H_2.
(iii) For any reaction greater the value of equilibrium constant greater is the extent of completion of reaction.
(iv) For the reaction, $N_2O_4 \rightleftharpoons 2NO_2$, $K_c = K_p/RT$.
(v) For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the degree of dissociation of PCl_5 increases with increase in pressure.
(vi)The value of equilibrium constant does not depend upon pressure.
How many of these statements are correct :

(1) 2

(2) 3

(3) 4

(4) 5

14) **Assertion :** Introduction of inert gas in any gaseous reaction at equilibrium, keeping pressure constant, always shifts the equilibrium in forward direction.
Reason : Due to introduction of inert gas at constant pressure, the partial pressures of various reactants and products decrease.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true but Reason is false

(4) Assertion is False but reason is true

15) One mole of PCl_5 is heated in a closed container of one litre capacity. At equilibrium, 20% PCl_5 is not dissociated. What should be the value of K_c ?

(1) $(3.2)^{-1}$

(2) 3.2

(3) 2.4

(4) 42

16) $AB_3(g)$ is dissociates as $AB_3(g) \rightleftharpoons AB_2(g) + \frac{1}{2}B_2(g)$. When the initial pressure of AB_3 is 800 torr and the total pressure developed at equilibrium is 900 torr. What percentage of $AB_3(g)$ is dissociated?

(1) 10%

(2) 20%

(3) 25%

(4) 30%

17) For the reaction $2A \rightleftharpoons B + C$, $K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is :
$[A] = [B] = [C] = 2 \times 10^{-3}$ M.
Then, which of the following is correct ?

(1) Reaction is at equilibrium.

(2) Reaction has a tendency to go in forward direction.

(3) Reaction has a tendency to go in backward direction.

(4) Reaction has gone to completion in forward direction.

18) In which of the following equilibria, K_p and K_c are **NOT** equal ?

(1) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

(2) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

(3) $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$

(4) $2BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}$

19) **Assertion :-** For the physical equilibrium $H_2O(s) \rightleftharpoons H_2O(\square)$ on increasing temperature and increasing pressure more water will form.
Reason :- Since forward reaction is endothermic in nature and density of water is less than that of the density of ice.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

20) Which of the following statement is incorrect ?

(1) On addition of catalyst the equilibrium constant value is not affected.

(2) Equilibrium constant for a reaction with negative ΔH value decreases as the temperature increases.

(3) In equilibrium mixture of ice and water kept in perfectly insulated flask, mass of ice and water does not change with time

(4) All the measurable properties of the system are variable at equilibrium state

21) At constant temperature, the equilibrium constant (K_p) for the decomposition reaction $N_2O_4 \rightleftharpoons$

$2NO_2$ is expressed by $K_P = \frac{\left(4x^2P\right)}{\left(1-x^2\right)}$, where P = Pressure, x = extent of decomposition. Which one of the following statement is true ?

(1) K_P increases with increase of P

(2) K_P increases with increase of x

(3) K_P increases with decrease of x

(4) K_P remains constant with change in P and x

22) Solubility of AgBr will be minimum in :-

(1) Pure water

(2) 0.1 M $CaBr_2$

(3) 0.1 M NaBr

(4) 0.1 M $AgNO_3$

23) Consider :-
(a) $FeCl_3$ in water - Basic
(b) NH_4Cl in water - Acidic
(c) Ammonium acetate in water - Acidic
(d) Na_2CO_3 in water - Basic
Which is/are not correctly matched:-

(1) b and d

(2) b only

(3) a and c

(4) d only

24) Which of the following statement (s) is/are correct:-
(A) larger the value of Ka, stronger is the acid.
(B) Conjugate acid of H_2O is OH^-.
(C) pH of 10^{-8}M HCL(aq.) solution is 8.
(D) K_w of water increases with increase in temperature.
Choose the correct answer from the options given below:

(1) A, C, D

(2) A, D

(3) A, B,C, D

(4) A, B, D

25) The rapid change of pH near the end point of an acid–base titration is the basis of indicator detection. pH of the solution is related to ratio of the concentrations of the conjugate acid (HIn) and base (In^-) forms of the indicator by the expression :-

(1) $\log\frac{[HIn]}{[In^-]} = pK_{In} - pH$

(2) $\log \frac{[HIn]}{[In^-]} = pH - pK_{In}$

(3) $\log \frac{[In^-]}{[HIn]} = pH + pK_{In}$

(4) $\log \frac{[In^-]}{[HIn]} = pK_{In} - pH$

26) Phosphoric acid ionizes in three steps with their ionization constant values K_{a_1}, K_{a_2} and K_{a_3}, respectively,
While K is the overall ionization constant.
Which of the following statements are true ?
A. $\log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$
B. H_3PO_4 is a stronger acid than $H_2PO_4^-$ and HPO_4^{2-}
C. $K_{a_1} > K_{a_2} > K_{a_3}$
D. $K_{a_1} = \frac{K_{a_3} + K_{a_2}}{2}$
Choose the **correct** answer from the options given below:

(1) A and B only

(2) A and C only

(3) B, C and D only

(4) A, B and C only

27) Calculate the molar solubility of $Ni(OH)_2$ in 0.10 M NaOH. The solubility product of $Ni(OH)_2$ is 2×10^{-15}.

(1) 2×10^{-13} M

(2) 3.2×10^{-8} M

(3) 5×10^{-5} M

(4) 3×10^{-17} M

28) If 100 mL of pH = 3 and 400 mL of pH = 3 is mixed, what will be the pH of the mixture

(1) 3.2

(2) 3.0

(3) 3.5

(4) 2.8

29) The degree of hydrolysis of which of the following salt is independent of the concentration of salt solution?

(1) CH_3COONa

(2) NH_4Cl

(3) CH_3COONH_4

(4) NaCl

30) 10 ml of $\frac{M}{200}$ H_2SO_4 is mixed with 40 ml of $\frac{M}{100}$ HCl and 50 ml of 0.004 M $Ba(OH)_2$. The pH of the resulting solution is -

(1) 2
(2) 3
(3) 4
(4) 1

31) The pK_a of HCN is 9.30. The pH of a solution prepared by mixing 2.5 moles of KCN and 2.5 moles of HCN in water and making up the total volume of 500 mL is :

(1) 9.30
(2) 7.30
(3) 10.30
(4) 8.30

32) A buffer solution is prepared by mixing 0.2 moles of NH_3 and 0.1 moles of NH_4Cl is l liter solution. the $(K_b)_{NH_3} = 1.8 \times 10^{-5}$. What is the pH of buffer solution

(1) 8.95
(2) 9.25
(3) 9.45
(4) 9.54

33) The pH of 0.01 M sodium acetate solution is : $[K_a(CH_3COOH)] = 2 \times 10^{-5}$

(1) 7.25
(2) 6.5
(3) 8.05
(4) 8.35

34) An acid solution of pH = 6 is diluted 100 times. The pH of solution becomes

(1) 6.98
(2) 6
(3) 4
(4) 8

35) When 0.4gm of NaOH is dissolved in 10 lit of solution, the pH at 90°C is :-

(1) 3
(2) 12
(3) 11
(4) 9

36) At 25°C, pH of water is 7. If water is heated at 50°C which one of the following is true -

(1) pH will decrease
(2) pH will increase
(3) pH will remain constant
(4) (H^+) will increase but (OH^-) will decrease

37) **Assertion :** pH of hydrochloric acid solution is less than that of acetic acid solution of the same concentration.
Reason : In equimolar solutions the number of titrable protons present in hydrochloric acid is less than that present in acetic acid.

(1) Both Assertion & Reason are True and the Reason is the correct explanation of the Assertion
(2) Both Assertion & Reason are True but the Reason is not the correct explanation of the Assertion
(3) Assertion is True statement but Reason is false
(4) Both Assertion and Reason are False statements

38) Which of the following is a buffer solution?

(1) 100 ml, 0.1 N CH_3COOH + 50 ml, 0.1 N NaOH
(2) 100 ml, 0.1 N HCl + 100 ml, 0.1 N NaOH
(3) 50 ml, 0.1 N CH_3COOH + 100 ml, 0.1 N NaOH
(4) 100 ml, 0.1 N HCl + 50 ml, 0.1 N NaOH

39) When sodium acetate (CH_3COONa) is added to aqueous solution of acetic acid (CH_3COOH), the -

(1) The pH value becomes zero
(2) pH value remains unchanged
(3) pH value decreases
(4) pH value increases

40) Which of the following solution will have maximum pH.

(1) 0.1 M HCOOK
(2) 0.1 M $MgCl_2$
(3) 0.1 M $NaNO_3$
(4) 0.1 M $FeSO_4$

41) For a weak acid HA, the percentage of ionisation is 1% at equilibrium. If the initial concentration of acid is 0.02M, then the correct option for its K_a at the same temperature is :-

(1) 2×10^{-6}
(2) 2×10^{-4}
(3) 2×10^{-3}
(4) 2×10^{-2}

42) pH of 0.1 M aqueous sodium iodide (NaI) solution will be :-

(1) 11.0
(2) 0.0
(3) 5.0
(4) 7.0

43) Degree of dissociation of 0.05 M NH_3 solution of pH = 11 is at 25°C :-

(1) 2%
(2) 3%
(3) 4%
(4) 5%

44) Given below are two statements :
Statement-I : Solubility of AgBr decreases in presence of Ammonia.
Statement-II : AgBr forms a complex $[Ag(NH_3)_2]^+$ in presence of Ammonia.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement-I & statement-II are incorrect
(2) Statement-I & statement-II are correct
(3) Statement-I is incorrect but statement-II is correct
(4) Statement-I is correct but statement-II is incorrect

45) An acidic buffer is prepared by mixing :

(1) weak acid and it's salt with strong base
(2) equal volumes of equimolar solutions of weak acid and weak base
(3) strong acid and it's salt with strong base
(4) strong acid and it's salt with weak base
(The pK_a of acid = pK_b of the base)

BIOLOGY

1) While studying moth populations in an industrial area, a researcher observed that dark-coloured moths were more abundant on tree trunks covered with soot, whereas light-coloured moths were rarely seen. In nearby non-industrial areas, light-coloured moths were more common. What evolutionary phenomenon does this observation support?

(1) Genetic drift
(2) Industrial melanism
(3) Adaptive radiation
(4) Artificial selection

2) Fossil records provide important evidence for evolution. Fossils are the preserved remains or

impressions of organisms that lived in the past. The age of fossils can be determined by studying the rock layers in which they are found. Older fossils are generally found in deeper layers, while newer fossils occur in upper layers.
Given below are two statements about fossils and evolution.
Statement-I: Fossils found in deeper rock layers represent organisms that lived earlier in geological history.
Statement-II: Fossils provide evidence that different life forms existed at different periods of time and many of them became extinct.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

3) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Homologous organs in different organisms provide strong evidence for evolution, as they share a common structural plan but are modified to perform different functions in different organisms.
Reason (R): Homologous organs arise due to divergent evolution, where organisms with a common ancestor evolve different adaptive features in response to different environmental conditions.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

4) Given below are two statements:
Statement I: Oparin of Russia and Haldane of England proposed that the first form of life could have come from pre-existing non-living organic molecules and that the formation of life was preceded by chemical evolution.
Statement II: In 1953, S. L. Miller created similar conditions of early earth in a laboratory and observed the formation of amino acids.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

5) Which of the following statements are correct?
A. Homologous organs are the result of divergent evolution.
B. Analogous organs are the result of convergent evolution.
C. Adaptive radiation is the process of evolution of different species in a given geographical area starting from a point and radiating to other areas of geography.

D. Wings of butterfly and wings of birds are homologous organs.
E. Australian marsupials represent adaptive radiation.
Choose the correct answer:

(1) A, B, C only
(2) B, C, D, E only
(3) A, B, C, E only
(4) A, D, E only

6) Evolutionary biology mainly deals with the study of which of the following aspects of life on Earth?

(1) Structural diversity of present-day organisms
(2) History of life forms in relation to earth
(3) Classification of living organisms
(4) Genetic composition of modern populations

7) The Big Bang theory primarily explains the origin of which of the following?

(1) Life on Earth
(2) Organic molecules
(3) Universe and its expansion
(4) First cellular organisms

8) Which of the following is an incorrectly matched pair?

(1) Louis Pasteur - Disproved spontaneous generation
(2) Oparin and Haldane - Chemical evolution theory
(3) S.L. Miller - Formation of amino acids in laboratory
(4) Charles Darwin - Theory of spontaneous generation

9) The early Earth atmosphere, mainly contained:

(1) Oxygen, nitrogen and ozone
(2) Methane, ammonia, hydrogen and water vapour
(3) Carbon monoxide and sulphur dioxide
(4) Oxygen and carbon dioxide only

10) Fill in the blanks correctly:
The first non-cellular forms of life are believed to have originated about _____ billion years ago, while the first cellular forms appeared about _____ million years ago.

(1) 3; 2000
(2) 2; 3500
(3) 4; 1500
(4) 1; 500

11) According to Darwinian theory, fitness of an organism refers to its:

(1) Physical strength and longevity
(2) Ability to adapt morphologically
(3) Reproductive success under given conditions
(4) Resistance to environmental stress

12) Which of the following statements regarding fossils is correct?

(1) Fossils are found only in igneous rocks
(2) Fossils indicate that all life forms appeared simultaneously
(3) Different sedimentary layers contain fossils of different ages
(4) Fossils are direct evidence of spontaneous generation

13) Which of the following sets represents homologous structures correctly?

(1) Wings of butterfly and bird
(2) Eye of octopus and mammals
(3) Forelimbs of whale, bat and human
(4) Flippers of penguin and dolphin

14) Which statement is correct?

(1) Homology results from convergent evolution
(2) Analogy indicates common ancestry
(3) Homology is based on divergent evolution
(4) Analogy involves similar origin but different functions

15) Industrial melanism observed in peppered moths demonstrates:

(1) Genetic drift
(2) Mutation pressure
(3) Natural selection favouring better camouflage
(4) Artificial selection by humans

16) Adaptive radiation refers to:

(1) Evolution of similar traits in unrelated organisms
(2) Gradual extinction of species from a region
(3) Evolution of multiple species from a common ancestor in a region
(4) Sudden appearance of mutations causing speciation

17) According to Hardy-Weinberg principle, a population is said to be in genetic equilibrium when:

(1) Mutation rate is high
(2) Allele frequencies remain constant over generations

(3) Natural selection is actively operating
(4) Speciation is occurring

18) Which of the following factors does NOT disturb Hardy-Weinberg equilibrium?

(1) Gene migration
(2) Genetic drift
(3) Random mating
(4) Natural selection

19) Which correctly represents the evolutionary sequence of early vertebrates?

(1) Reptiles - Amphibians - Fish
(2) Fish - Amphibians - Reptiles
(3) Amphibians - Fish - Reptiles
(4) Fish - Reptiles - Amphibians

20) Which of the following statements about human evolution is correct?

(1) *Homo habilis* had the largest brain size
(2) *Homo habilis* were strictly carnivorous
(3) *Homo* erectus showed increased brain capacity and upright pasture
(4) Neanderthals lacked cultural practices

21) Embryos of advanced species pass through stages represented by adult organisms of more primitive species. The given statement is related with :-

(1) Baer's law
(2) Dollo's law
(3) Mutation theory
(4) Recapitulation theory

22) _____ is the end result of the ability to adapt and get selected by nature. Complete the statement with appropriate word :-

(1) Sexual selection
(2) Inheritance
(3) Fitness
(4) Variation

23) Variation in gene frequencies within populations can occur by chance rather than by natural selection. This is referred to as :-

(1) Gene migration
(2) Gene flow
(3) Genetic drift

(4) Genetic recombination

24) Theory of inheritance of acquired characters was given by :-

(1) Wallace
(2) Lamarck
(3) Darwin
(4) De vries

25) _____ called single step large mutation as saltation :

(1) Hugo de vries
(2) Darwin
(3) Lamarck
(4) Mendel

26) Jawless fishes probably evolved around :-

(1) 500 mya
(2) 350 mya
(3) 320 mya
(4) 65 mya

27) When natural selection leads to stabilization, then in frequency distribution curve of a population:

(1) more individuals acquire mean character value
(2) more individuals acquire value other than mean character value
(3) more individuals acquire peripheral character value at both ends of the curve
(4) individuals acquire all character values equally

28) Which of the following had smallest brain capacity?

(1) Modern man
(2) Java man
(3) Neanderthal man
(4) *Homo habilis*

29) **Assertion :-** S.L. Miller with his experiment proved Big-bang Theory.
Reason :- Big-bang Theory explain origin of Life.

(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion
(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion
(3) Assertion is correct & Reason is incorrect
(4) Both Assertion & Reason are incorrect

30) **Assertion :-** Homology indicates common ancestry.
Reason :- The same structure developed along different direction due to adaptations to different needs is homology.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

31) **Assertion :** Pouched mammals survived in Australia.
Reason : They had lack of competition from any other mammals in Australia.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion & reason are false.

32) **Assertion :** An analysis of meteorite supports chemical evolution.
Reason : Analysis of meteorite content revealed similar compounds indicating that similar process are occurring elsewhere in space.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

33) **Statement-I :** Thorns of *Bougainvillea* and tendrils of *Cucurbita* represents homology.
Statement-II : Homology does not indicate common ancestry.

(1) Both Statement-I and II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

34) **Statement-1 :** *Dryopithecus* was more man like.
Statement-2 : *Ramapithecus* was more ape like.

(1) Both statement are correct
(2) Statement-1 is correct, Statement-2 is incorrect
(3) Statement-1 is incorrect, Statement-2 is correct
(4) Both Statement are incorrect

35) **Statement-I :** Branching descent and natural selection are two key concept of Darwinian Theory of Evolution.
Statement-II : Natural selection is based on certain observation which are factual.

(1) Statement-I correct & Statement-II incorrect.

(2) Statement-I incorrect & Statement-II correct.

(3) Statement-I correct & Statement-II correct.

(4) Statement-I incorrect & Statement-II incorrect.

36) **Statement-I :** In 1938 a fish caught in South Africa happened to be a coelacanth which was thought to be extinct.
Statement-II : The amphibian evolved into reptile. They lay thick shelled egg which do not dry up in sun unlike those of amphibian.

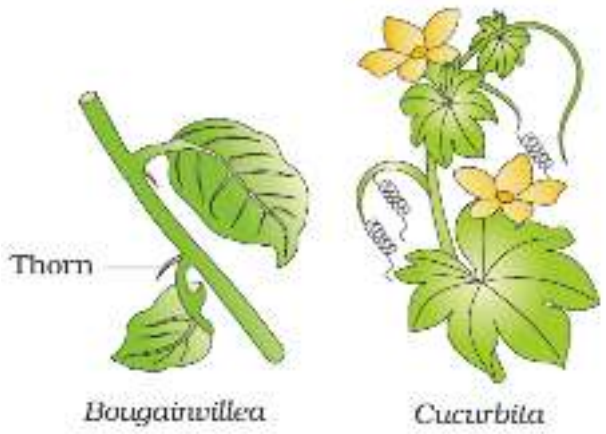
(1) Statement-I correct & Statement-II incorrect.

(2) Statement-I incorrect & Statement-II correct.

(3) Statement-I correct & Statement-II correct.

(4) Statement-I incorrect & Statement-II incorrect.

37) Thorn of *Bougainvillea* and tendril of *Cucurbita* represents :



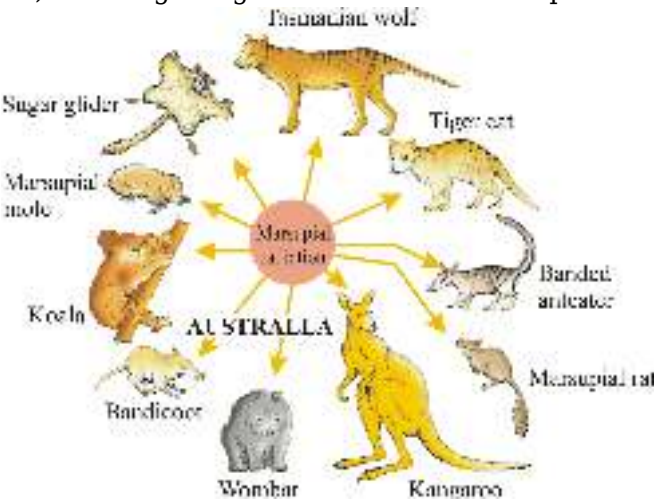
(1) Convergent evolution

(2) Analogous organs

(3) Divergent evolution

(4) Parallel evolution

38) The diagram given below is of marsupials of Australia, provides an example of :-

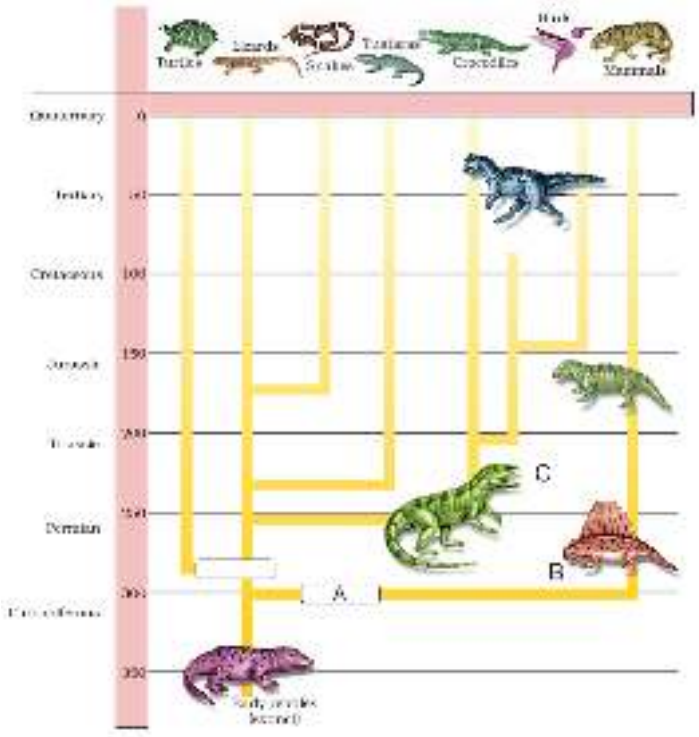


(1) Adaptive convergence

(2) Parallel evolution

(3) Convergent evolution

(4) Adaptive radiation



39) Identify A,B and C from above figure :-

(1) A-sauropsid, B-therapsid, C-pelycosaurs

(2) A-sauropsid, B-pelycosaurs, C-therapsid

(3) A-sauropsid, B-thecodonts, C-pelycosaurs

(4) A-synapsids, B-pelycosaurs, C-thecodonts

40) Identify the correct match from column I and column II :–

Column-I	Column-II
(A) Origin of Universe	(i) 4 bya
(B) Origin of earth	(ii) 4.5 bya
(C) Origin of first cellular form of life	(iii) 3 bya
(D) Origin of first non-cellular form of life	(iv) 13.8 bya
(E) Life appeared	(v) 2 bya

(1) A – iv, B – i, C – iii, D – ii, E – v

(2) A – iv, B – i, C – v, D – ii, E – iii

(3) A – iv, B – ii, C – v, D – iii, E – i

(4) A – iv, B – ii, C – iii, D – v, E – i

41) Which of the following option is **incorrect** ?

(1)	Theory of special creation	Diversity was same and will be same in future.
(2)	Theory of Biogenesis	New life can be originated on earth by decaying and rotting matter.
(3)	Saltation	Single step large mutation
(4)	Genetic drift	May result from founder effect

(1) 1

(2) 2

(3) 3

(4) 4

42) Read the following statements and find out the true (T) or false (F) statements :-
(A) Study of history of life forms on earth is known as evolutionary biology.
(B) The universe is very old-almost 13.8 billion years old.
(C) Earth was supposed to have been formed about 4.0 billion years back.
(D) Spontaneous generation theory attempts to explain to us the origin of universe.
Options :

	A	**B**	**C**	**D**
(1)	T	T	F	T
(2)	T	F	T	F
(3)	T	T	F	F
(4)	F	F	T	F

(1) 1

(2) 2

(3) 3

(4) 4

43) Which of the following statements are true for evolution?
(a) *Homo habilis* probably ate meat.
(b) *Homo erectus* did not eat meat.
(c) *Dryopithecus* were more ape-like, while *Ramapithecus* were more man-like.
(d) Agriculture and human settlements started around 18000 years back.
(e) *Australopithecus* lived in East African grasslands.
Choose the most appropriate answer from the options given below:

(1) (c) only

(2) (c), (d) and (e) only

(3) (c) and (e) only

(4) (a), (b) and (d) only

44) How many of the following set of mammals showing convergent evolution ?
Mouse-marsupial mouse, Numbat-spotted cuscus, mole-marsupial mole, Lemur-Anteater, flying squirrel-flying phalanger, bob cat-koala.

(1) 5

(2) 3

(3) 4

(4) 2

45) Arrange the following men according to their increasing cranial capacity :
(a) *Homo habilis*
(b) *Homo erectus*
(c) *Homo sapiens*

(1) a, b, c

(2) b, c, a

(3) c, a, b

(4) c, b, a

46) Metagenesis means :-

(1) Polyp produce medusae asexually and medusae produce polyp sexually

(2) Polyp produce medusae sexually and medusae produce polyp sexually

(3) Polyp produce medusae asexually and medusae produce polyp asexually.

(4) Both (1) & (2)

47) Muscular pharynx is a characteristic feature of :

(1) *Taenia*

(2) *Hydra*

(3) *Ascaris*

(4) *Sycon*

48) In which animal polyp and medusa forms occurs ?

(1) Hydra

(2) *Obelia*

(3) *Aurelia*

(4) Sea-pen

49) Ladder like nervous system is a feature of :

(1) Coelenterata

(2) Ctenophora

(3) Platyhelminthes

(4) Porifera

50) Phylum of animals having single opening to the outside that serves both as mouth and anus is :-

(1) Platyhelminthes

(2) Mollusca
(3) Echinodermata
(4) Hemichordata

51) Water transport system is characteristic feature of :-

(1) Echinodermata
(2) Porifera
(3) Cnidaria
(4) Ctenophora

52) 'Corals' belongs to phylum :-

(1) Hemichordata
(2) Mollusca
(3) Cnidaria
(4) Porifera

53) The space between body wall and alimentary canal lined by mesoderm is called :-

(1) Acoelom
(2) Pseudocoelom
(3) Coelom
(4) None of these

54) See the following diagrams carefully,these animals are :



(1) *Sycon, Leucosolenia*
(2) *Hydra, Aurelia*
(3) *Nereis, Hirudinaria*
(4) *Fasciola, Taenia*

55) Observe the given figure : Select correct answer for it :



(1) It is fresh water sponge
(2) Animal body is porous
(3) It has radial symmetry
(4) Both 1 & 2

56) Chitinuous exoskeleton is found in which phylum ?

(1) Annelida
(2) Mollusca
(3) Arthropoda
(4) Echinodermata

57) Identify the correct matching pair?

(1) Vector–Silkworm
(2) Economically important insect –*Anopheles*
(3) Gregarious pest – *Culex*
(4) Living fossil – King crab

58) Water vascular system is the most distinctive feature of phylum :-

(1) Mollusca
(2) Echinodermata
(3) Porifera
(4) Arthropoda

59) *Balanoglossus* is an example of :-

(1) Echinodermata
(2) Hemichordata
(3) Arthropoda
(4) Chordata

60) The fertilisation is _____ and development is _____ in sponges.

(1) External, Indirect
(2) External, Direct
(3) Internal, Indirect
(4) Internal, Direct

61) In which phylum Nephridia helps in osmoregulation and excretion ?

(1) Annelida
(2) Mollusca
(3) Arthropoda
(4) Porifera

62) Character which is not shared by members of Aschelminthes and Annelida :-

(1) Organ system level of organisation
(2) Bilaterally symmetrical body
(3) Triploblastic
(4) Metameric segmentation

63) ***Assertion*** :- '*Obelia*' shows the metagenesis.
Reason :- In the life cycle of *Obelia*, polyp and medusa forms are alternately repeated.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

64) **Assertion :** Hemichordates are considered as a subphylum under phylum chordata.
Reason : Hemichordates have a rudimentary structure in collar called stomochord, similar to notochord.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion is false but the Reason is True.

65) Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Aschelminthes are called as round worms.
Reason (R) : Their body is circular in cross section.
In the light of the above statements, choose the correct answer from the options given below –

(1) Both A and R are correct but R is not the correct explanation of A
(2) A is correct but R is not correct
(3) A is not correct but R is correct
(4) Both A and R are correct and R is the correct explanation of A

66) **Assertion :** Platyhelminthes have incomplete digestive system.
Reason : Their digestive system has only a single opening to the outside of the body that serves as both mouth and anus.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

67) Given below are two statements –
Statement-I : In Aschelminthes alimentary canal is complete with a well developed muscular pharynx.

Statement-II : Male *Ascaris* is longer than female.
In light of the above statements, choose the correct answer from the option given below.

(1) Both statements-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect
(3) Statement-I is incorrect but Statement-II is correct
(4) Both statements-I and Statement-II are correct.

68) **Statement-I :** *Balanoglossus* is unisexual.
Statement-II : Fertilization is external with direct development.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

69) **Statement-I :** All members of phylum porifera are asymmetrical and marine.
Statement-II : Echinodermata exhibits radial or bilateral symmetry depending on the stage.

(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect
(3) Only statement-I is correct.
(4) Only statement-II is correct.

70) **Statement-I :** Ctenophora show bioluminescence property and only reproduce sexually.
Statement-II : *Planaria* possess high regeneration capacity and show internal fertilization.

(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect
(3) Only statement-I is correct.
(4) Only statement-II is correct.

71) Match column 'A' and column 'B' & choose the correct option

	Column'A'		**Column 'B'**
a	*Sycon*	i	Sea fan
b	*Physalia*	ii	Scypha
c	*Gorgonia*	iii	Sea anemone
d	*Adamsia*	iv	Porturguese man of war

(1) a-ii, b-iv, c-iii, d-i
(2) a-i, b-ii, c-iii, d-iv
(3) a-ii, b-iv, c-i, d-iii
(4) a-iii, b-i, c-ii, d-iv

72)

Match the given columns :

(A) *Echinus* (i) Brittle star
(B) *Antedon* (ii) Sea-urchin
(C) *Ophiura* (iii) Sea-lily

(D) *Cucumaria* (iv) Sea-cucumber

(1) A-(i), B-(ii), C-(iii), D-(iv)

(2) A-(ii), B-(iii), C-(i), D-(iv)

(3) A-(ii), B-(iv), C-(iii), D-(iv)

(4) A-(i), B-(iii), C-(ii), D-(iv)

73) Match the column-I and II and select the correct option?

	Column-I		**Column-II**
(A)	Prawn	(i)	Trachea
(B)	Scorpion	(ii)	Moist cuticle
(C)	Cockroach	(iii)	Book lungs
(D)	Earthworm	(iv)	Book gills
		(v)	Gills

	A	B	C	D
(1)	v	iii	i	iv
(2)	iv	v	i	ii
(3)	v	iii	i	ii
(4)	v	i	ii	iii

(1) 1

(2) 2

(3) 3

(4) 4

74)

Match list 'A' and list 'B' correctly and find your answer from the code given below :-

List-A (Animal)		**List-B (Common name)**	
(i)	*Adamsia*	(A)	Brittle star
(ii)	*Ophiura*	(B)	Squid
(iii)	*Limulus*	(C)	Sea hare
(iv)	*Aplysia*	(D)	Sea anemone
(v)	*Loligo*	(E)	King crab

(1) (i)-D, (ii)-C, (iii)-E, (iv)-A, (v)-B

(2) (i)-C, (ii)-D, (iii)-E, (iv)-B, (v)-A

(3) (i)-D, (ii)-A, (iii)-E, (iv)-C, (v)-B

(4) (i)-C, (ii)-D, (iii)-B, (iv)-A, (v)-E

75)

Match the column-I with column-II

	Column-I (Phylum)		Column-II (Level of organisation)
(A)	Coelenterata	(i)	Cellular
(B)	Aschelminthes	(ii)	Tissue
(C)	Porifera	(iii)	Organ
(D)	Platyhelminthes	(iv)	Organ system

(1) A-iv,B-iii, C-ii, D-i

(2) A-ii,B-iv, C-iii, D-i

(3) A-ii,B-iv, C-i, D-iii

(4) A-i,B-ii, C-iii, D-iv

76) Choose the correct option in which blind sac type body plan is present :-

(1) Porifera, Coelentereta, Ctenophora

(2) Coelentereta, Ctenophora, Platyhelminthes

(3) Aschelminthes, Annelida, Arthropoda

(4) Aschelminthes to Chordata

77) In the given list, segmentation is present in how many of the following ?
Annelida, Arthropoda, Chordata, Mollusca, Aschelminthes

(1) 4

(2) 3

(3) 2

(4) 1

78) Which of the following options is correct about the figure given below along with its phylum?



(1) Cnidoblast cell - Coelenterata

(2) Flame cell - Platyhelminthes

(3) Lasso cell - Ctenophora

(4) Trichocyst - Protozoa

79) Choose the correct match from column-I, II and III :-

	Column-I		Column-II		Column-III
A		i	Platyhelminthes	a	Tissue level of organisation
B		ii	Aschelminthes	b	Organ system level of organisation
C		iii	Porifera	c	Organ/organ system level of organisation
D		iv	Coelenterata	d	Cellular level of organisation

(1) A-i-a, B-ii-b, C-iii-c, D-iv-d

(2) A-ii-b, B-iv-a, C-i-c, D-iii-d

(3) A-iii-c, B-i-d, C-ii-a, D-iv-a

(4) A-iv-d, B-iii-c, C-iv-b, D-i-b

80) Which of the following option is correct and without exception ?

(1) Fertilization is internal and development is indirect- *Scypha*, *Ctenoplana*

(2) Flame cells help in osmoregulation and alimentary canal present-*Fasciola, Taenia*

(3) Body is segmented and performs locomotion with the help of parapodia- *Nereis, Pheretima*

(4) Body is covered by mantle layer and gills are present - *Pinctada, Pila*

81)

Read the following four statements (A-D) :-
(A) Water transport system is helpful in food gathering, respiration and removal of wastes.
(B) All members of phylum porifera are marine and symmetrical.
(C) Sponges are primitive multicellular animals and have cellular level of organisation.
(D) In sponges, fertilisation is external and development is direct.

How many of the above statements are incorrect?

(1) One

(2) Four

(3) Two

(4) Three

82) Which of the following statement is/are correct regarding phylum coelenterata ? (A) They are aquatic, mostly marine, sessile or free swimming, radially symmetrical animals.
(B) They have a central gastro-vascular cavity.
(C) Digestion is extracellular and intracellular.
(D) Examples are *Sycon*, *Spongilla and Euspongia*

(1) A and B

(2) A and D

(3) A, B and C

(4) All of these

83) Read the following statements A-D :-
(A) Members are worm like and marine
(B) Body divided into proboscis, collar and trunk
(C) Respiration by gills
(D) Flame cells are excretory organs
How many statements are incorrect for *Balanoglossus*.

(1) One

(2) Two

(3) Three

(4) Four

84) Which of the following is not a character of phylum echinoder mata :-
(i) They have water vascular system
(ii) All are sessile
(iii) They have endoskeleton
(iv) They have open circulatory system
Options:-

(1) (i) & (iii)

(2) (ii) & (iii)

(3) Only (ii)

(4) Only (iv)

85) Read the following features :-
(i) Commonly known as "Round worm"
(ii) Animals are bilaterally symmetrical and triploblastic
(iii) Exclusively parasite on animals
(iv) Pseudocoelomate animals having organ-system level of body organisation
(v) Body shows segmentation
(vi) Alimentary canal complete with well developed muscular pharynx
(vii) Animals are bisexual and shows sexual dimorphism
(viii) Fertilization internal and direct or indirect development.

Out of these how many features are not exhibited by members of Aschelminthes ?

(1) Two

(2) Three

(3) Four

(4) One

86) Read the following statements A – D :-
(A) They have orgen level of organisation
(B) Bilateral symmetrical, triploblastic and acoelomate
(C) Hooks and suckers in parasitic form
(D) Fertilization internal
Above statements are true for which animal.

(1) *Ascaris*

(2) *Hydra*

(3) *Spongilla*

(4) *Taenia*

87) Which is not a sponges?

(1) *Sycon*

(2) *Spongilla*

(3) Scypha

(4) Sea anemone

88)

Match Column-I with Column-II on basis of excretory organs.

	Column-I		**Column-II**
(a)		(i)	Malpighian tubules
(b)		(ii)	Coxal Gland
(c)		(iii)	Proboscis Gland

(d)		(iv)	Nephridia

(1) a-i, b-ii, c-iii, d-iv

(2) a-iii, b-i, c-iv, d-ii

(3) a-iv, b-iii, c-ii, d-i

(4) a-ii, b-i, c-iv, d-iii

89) How many animals are related with echinodermata
Asterias, *Echinus*, *Sea lily*, *Ophiura*, *Cucumaria*, *Saccoglossus*, *Cuttle fish*, *Sea hare*, *Culex*, *Chiton*

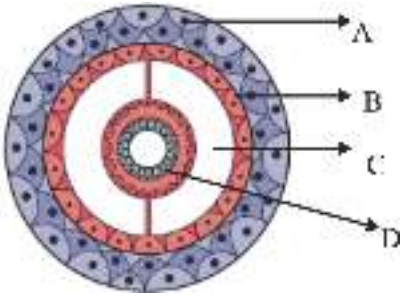
(1) 5

(2) 6

(3) 4

(4) 7

90) Given below is the diagrammatic sketch of the sectional view of an animal. Identify the parts labelled as A, B, C and D and select the right option about them :-



	Part-A	**Part-B**	**Part-C**	**Part-D**
1	Ectoderm	Mesoderm	Ectoderm	Coelom
2	Mesoderm	Ectoderm	Coelom	Ectoderm
3	Ectoderm	Mesoderm	Coelom	Endoderm
4	Ectoderm	Coelom	Mesoderm	Ectoderm

(1) 1

(2) 2

(3) 3

(4) 4.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	4	3	1	3	3	2	3	2	4	3	4	4	4	3	1	2	3	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	3	2	3	1	1	3	2	4	3	1	1	3	1	1	1	2	3	1	4
Q.	41	42	43	44	45															
A.	3	3	1	4	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	4	4	4	3	3	1	2	2	3	2	4	3	4	2	3	3	1	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	3	2	1	4	1	2	3	2	1	4	4	1	4	1	3	1	4	1
Q.	86	87	88	89	90															
A.	1	4	1	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	1	1	3	2	3	4	2	1	3	3	3	3	3	3	2	3	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	3	3	2	1	2	1	4	4	1	1	1	2	4	3	3	3	4	4	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	3	2	1	1	3	2	3	1	2	3	3	3	4	3	4	2	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	4	4	4	4	1	2	2	4	1	3	2	3	3	3	2	2	1	2	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	1	3	2	4	4	2	1	3										

SOLUTIONS

PHYSICS

1) $v = at, \frac{ds}{dt} = at \Rightarrow ds = at\,dt$

$$(v_{avg})_{space} = \frac{\int v\,ds}{\int ds} = \frac{\int at\,at\,dt}{\int at\,dt} = \frac{a^2\frac{t^3}{3}}{\frac{at^2}{2}} = \frac{2at}{3}$$

$$(v_{avg})_{time} = \frac{\int v\,dt}{\int dt} = \frac{\int at\,dt}{\int dt} = \frac{a\frac{t^2}{2}}{t} = \frac{at}{2}$$

$$\frac{(v_{avg})_{space}}{(v_{avg})_{time}} = \frac{\frac{2at}{3}}{\frac{at}{2}} = \frac{4}{3}$$

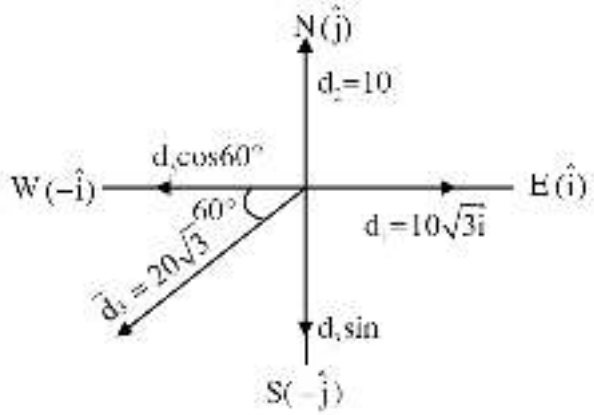
2) **Explain the Question :** The user wants to identify the correct relationship between the velocity and acceleration of a body in one-dimensional motion and how it affect its speed.
Concept : This question is based on speed changes based on the relative directions of velocity and acceleration.
Solution :
Speed increases when velocity and acceleration have the same sign. Speed decreases when they have opposite signs.

When velocity is negative and acceleration is positive, they are in opposite directions, causing the body to slow down.
Correct Given $v < 0$ and $a > 0$, the vectors for velocity and acceleration point in opposite directions. This opposition causes the magnitude of the velocity, which is speed, to decrease.
Final answer : 4

3)

Explanation the Question:
We have to calculate net displacement.
Concept:
This question is based on
$\vec{d}_{net} = \vec{d}_1 + \vec{d}_2 + \vec{d}_3 + ...$

Solution:



$\vec{d}_1 = 10\sqrt{3}\hat{i},\ \vec{d}_2 = 10\hat{j},\ d_3 = d_3\cos 60(-\hat{i}) + d_3\sin 60(-\hat{j})$

$d_3 = 20\sqrt{3} \times \frac{1}{2}(-\hat{i}) + 20\sqrt{3} \times \frac{\sqrt{3}}{2}(-\hat{j})$

$\vec{d} = \vec{d}_1 + \vec{d}_2 + \vec{d}_3$

$\vec{d} = 20(-\hat{j})$

so displacement will be 20'm', south

Final Answer: (3)

4)

Explain the Question : Asking about the ratio of the distance travelled during the 4th and 3rd second.

Concept : This question is based on motion with uniform acceleration.

Solution :

From the equation of displacement in n^{th} second.

$S_n = u + \frac{a}{2}(2n-1)$

$\frac{S_{4^{th}}}{S_{3^{rd}}} = \frac{\frac{a}{2}(2(4)-1)}{\frac{a}{2}(2(3)-1)}$

and given that u = 0 then

S_4 (Distance in the 4th second) :

$S_4 = \frac{a}{2}(2(4)-1) = \frac{a}{2}(8-1) = \frac{7a}{2}$

S_3 (Distance in the 3rd second) :

$S3 = \frac{a}{2}(2(3)-1) = \frac{a}{2}(6-1) = \frac{5a}{2}$

$\frac{S_4}{S_3} = \frac{\frac{7a}{2}}{\frac{5a}{2}} = \frac{7a}{2} \times \frac{2}{5a} = \frac{7}{5}$

Final answer : 1

5)

$x = \sqrt{V+1}$

$x^2 = V + 1$

$V = x^2 - 1$

$a = V\frac{dV}{dx} = (x^2 - 1)(2x)$

at x = 5

a = 24 × 10 = 240 unit

6) **Question Explanation:** The given velocity-time (v-t) graph shows a car moving along a straight road, where the velocity increases linearly from an initial velocity of 2 m/s at t = 0 s to a final velocity of 10 m/s at t = 10 s. Calculate the average velocity of the car during these first 10 seconds}.
Concept: Kinematics: Velocity-Time Graph Analysis
Solution :
Area of v-t graph gives displacement.

$$\text{Area} = \frac{1}{2} \times (\text{sum of parallel sides}) \times (\text{height})$$

$$\Delta x = \frac{1}{2} \times (2 + 10) \times 10$$

$$\Delta x = \frac{1}{2} \times 12 \times 10 = 60\ \text{m}$$

$$\text{Average Velocity} = \frac{\text{Total Displacement}}{\text{Total Time}}$$

$$\text{Average Velocity} = \frac{60\,\text{m}}{10\,\text{s}} = 6\,\text{m/s}$$

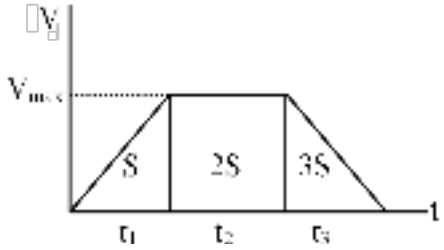
Final answer : Option (3)

7)

$$\sin\theta = \frac{v_R}{v} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

Angle with river flow = 90 + 30 = 120°

8) **Question Explanation :** A particle moves in a straight line some distance with uniform acceleration the with uniform speed after that uniform retardation we need to find out ratio of its average and maximum velocity.
Concept : Uniform accelerated motion.
Formula : $\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time}}$
Solution :



$$\frac{1}{2} t_1 V_{max} = S \quad \Big| \quad t_1 = \frac{2S}{V_{max}}$$

$$t_2 V_{max} = 2S \quad \Big| \quad t_2 = \frac{2S}{V_{max}}$$

$$\frac{1}{2} t_3 V_{max} = 3S \quad \Big| \quad t_3 = \frac{6S}{V_{max}}$$

$$V_{av} = \frac{6S}{t}$$

$V_{av} = \frac{6S}{10S} V_{max}$

$\frac{V_{av}}{V_{max}} = \frac{3}{5}$

Final Answer : Option (3)

9)

$\Rightarrow x = 4t - 8 + at^2 + 4a - 4at$

$\Rightarrow v = \frac{dx}{dt} = 4 + 2at - 4a$

$\Rightarrow$ acceleration $f = \frac{dv}{dt} = 2a$

10) $\vec{v} = \vec{u} + \vec{at}$

$\vec{v} = 3\hat{i} + 4\hat{j} + (0.4\hat{i} + 0.3\hat{j})$

$= 7\hat{i} + 7\hat{j}$

$|\vec{v}| = 7\sqrt{2}$ units

11)

$\Rightarrow h = \frac{u^2}{2g} = 5m \ \& \ t = \frac{u}{g} = \frac{10}{10} = 1 \sec$

$\Rightarrow s_y = u_y t + \frac{1}{2} a_y t^2$

$\Rightarrow h^1 = 10 \times \frac{1}{2} - \frac{1}{2} \times 10 \times \left(\frac{1}{2}\right)^2$

$= 5 - \frac{5}{4} = \frac{15}{4}$

$\Rightarrow h^1 = \frac{3}{4} \times 5 = \frac{3}{4} h$

12)

$\Rightarrow v^2 = u^2 + 2as \quad \Rightarrow v^2 = 2gh$

$\Rightarrow \left(\frac{v}{2}\right)^2 = 2g(H - h)$

$\Rightarrow \frac{2gH}{4} = 2gH - 2gh, \Rightarrow h = \frac{3}{4} H$

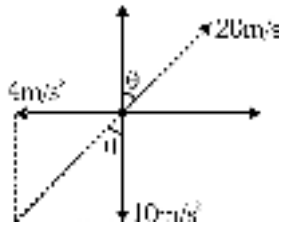
$\Rightarrow h = \frac{3}{4} \times 100 = 75$ m

13) $x = \alpha t^3$ and $y = \beta t^3$

$v_x = \frac{dx}{dt} = 3\alpha t^2$ and $v_y = \frac{dy}{dt} = 3\beta t^2$

$v = \sqrt{v_x^2 + v_y^2}$

$= \sqrt{(3\alpha t^2)^2 + (3\beta t^2)^2} = 3t^2\sqrt{\alpha^2 + \beta^2}$

14) 

The ball returns back to boy's hand only if the path of the ball is a straight line. Hence initial velocity and net acceleration must be along the same line. Hence

$\tan\theta = \frac{4}{10} \Rightarrow \theta = \tan^{-1}0.4$

15)

At t = 0,
$a_\perp = g\cos\theta$,

$R = \frac{v^2}{a_\perp} = \frac{u^2}{g\cos\theta}$

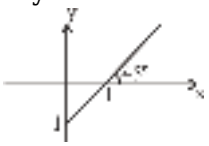
16)

As the trains are moving in the same direction. So the initial relative speed ($\upsilon_1 - \upsilon_2$) and by applying retardation final relative speed becomes zero.
From $\upsilon = u - at \Rightarrow$

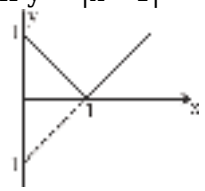
$0 = (\upsilon_1 - \upsilon_2) - at \Rightarrow t = \frac{\upsilon_1 - \upsilon_2}{a}$

17) $(v_y)_B = v \sin 60° = v\sqrt{3}/2$
$(v_y)_A = 2v \sin 30° = v$
$(v_y)_A > (v_y)_B$
So $T_A > T_B$

18) y = x–1, for x=0, y = –1 and for y = 0, x = 1
or $\frac{dy}{dx} = 1$
and $\frac{d^2y}{dx^2} = 0$
Thus, the graph is straight line having slope 1.
If y = x – 1



If y = |x – 1|



19) Maximum height = $\frac{1}{2} \times 110 \times 1000 = 55\,\text{km}$

20)

Equation of trajectory is

$y = x \tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

From given initial velocity

$u = \sqrt{u_x^2 + u_y^2}$

$u = \sqrt{1^2 + 2^2} = \sqrt{5}$

$\tan\theta = 2$

so $y = 2x - \frac{5x^2}{1} = 2x - 5x^2$

21) **Question explanation:** In this question, we are asked about the general relationship between two vectors $\vec{A}$ and $\vec{B}$ without any specific condition.

Concept: This question is based on type of vectors.

Solution: As Two arbitrary vectors $\vec{A}$ and $\vec{B}$ may on may not be parallel, equal or collinear. But any two vectors can always be contained in a plane ⇒ They are always coplanar

Final Answer: Option (4)

22) **Explain Question:**

The question asks for dot product of given vectors.

Concept:

Dot product of vectors.

Solution:

$(2\bar{A} + 3\bar{B}) \cdot (3\bar{A} - 2\bar{B})$

$=6A^2 - 4AB\cos\theta + 9\,BA\cos\theta - 6B^2$

$=6A^2 + 5\,AB\cos\theta - 6B^2$

Final answer: option (3)

23) **Explaining :** which of the following is not a vector from given options

Concept :

$\vec{A} + \vec{B} = \vec{R}$

$\vec{A} \cdot \vec{B} = C$

$\vec{A} - \vec{B} = \vec{R}$

$\vec{A} \times \vec{B} = \vec{C}$
Solution :
$\vec{A}.\vec{B} = AB\cos\theta$ (scalar)
Final Answer : (2)

24)

Explain Question : In this question we are asked to find a unit vector along the resultant of given $\vec{A}$ and $\vec{B}$.

Concept : This question is based on unit vector.

Solution :
$\vec{A} = 5\hat{i} - \hat{j} + 7\hat{k}$
$\vec{B} = -2\hat{i} + 3\hat{j} - \hat{k}$
The resultant vector $\vec{R} = \vec{A} + \vec{B}$
$\Rightarrow \vec{R} = (5\hat{i} - \hat{j} + 7\hat{k}) + (-2\hat{i} + 3\hat{j} - \hat{k})$
$\Rightarrow \vec{R} = 3\hat{i} + 2\hat{j} + 6\hat{k}$ .....(1)
A unit vector in the direction of vector $\vec{R}$ is
$\hat{R} = \dfrac{\vec{R}}{|\vec{R}|}$
$\Rightarrow |\vec{R}| = \sqrt{(3)^2 + (2)^2 + (6)^2}$ [from equation (1)]
$|\vec{R}| = \sqrt{9 + 4 + 36} = \sqrt{49} = 7$
Now, $\hat{R} = \dfrac{3\hat{i} + 2\hat{j} + 6\hat{k}}{7}$
Final answer : Option (3)

25) **Question Explanation:** Find the value of $|\hat{a} - \hat{b}|$ if $(\hat{a} + \hat{b}) = 2$

Concept: Addition and Subtraction of vectors

Solution:
Given here two unit vectors $\hat{a}$ and $\hat{b}$
$|\hat{a}| = \left|\hat{b}\right| = 1$
Therefore, Magnitude of the resultant vector
$R = \left|\hat{a} + \hat{b}\right| = \sqrt{(\hat{a}) + (\hat{b}) + 2|\hat{a}|(\hat{b})\cos\theta}$
$= 2 = \sqrt{(1)^2 + (1)^2 + 2.1.1.\cos\theta}$
$= 2 = \sqrt{2 + 2\cos\theta}$
Squaring both side we get
$\Rightarrow 4 = 2 + 2\cos\theta$
$\Rightarrow \cos\theta = 1$
$\Rightarrow \theta = 0^\circ (\because \cos 0^\circ = 1)$
$\left|\hat{a} - \hat{b}\right| = \sqrt{(\hat{a})^2 + (\hat{b})^2 - 2(\hat{a})(\hat{b})\cos\theta}$
$\left|\hat{a} - \hat{b}\right| = \sqrt{(1)^2 + (1)^2 - 2.1.1\cos 0^\circ} = 0$

Final Answer: Option (1)

26) **Explain question :** In this question, we have to determine which can not be the resultant of two given forces.

Concept : This question is based on resultant of two vectors (vector addition)

Solution : For the given two vectors
say, $F_1 = 10N$, $F_2 = 15$ N, the resultant 'R'
must satisfy the triangle inequality i.e.

$$|F_1 - F_2| \leqslant R \leqslant |F_1 + F_2|$$

$$\Rightarrow |10 - 15| \leqslant R \leqslant |10 + 15|$$

$5 \leq R \leq 25$
Any resultant outside this range is not possible
$\therefore$ 4N < 5 → it is not possible

Final answer : Option (1)

27) **Explain Question:**
To find angle between $\vec{A}$ and $\vec{B}$

Concept:
Addition and subtraction of vectors

Solution:

$$\sqrt{A^2 + B^2 + 2AB\cos\theta} = \frac{1}{2} \times \sqrt{A^2 + B^2 - AB\cos\theta}$$

$4(A^2 + B^2 + 2AB\cos\theta) = A^2 + B^2 - 2AB\cos\theta$
$3A^2 + 3B^2 + 10AB\cos\theta = 0$
$12B^2 + 3B^2 + 20B^2\cos\theta = 0$
$20B^2\cos\theta = -15B^2$

$$\cos\theta = -\frac{3}{4}$$

$$\theta = \cos^{-1}\left(-\frac{3}{4}\right)$$

Final answer: Option (3)

28) 

$S_{5th\ half} = S_{2.5} - S_2$

$$= \frac{1}{2} 2[(2.5)^2 - 2^2] = 2.25 \text{ m}$$

29) **Explain the Question :** To find induced current at t = 0.25 sec
Given $\phi = 6t^2 - 5t + 1$, R = 10 Ω
Concept : This question is based on

Induced current $I = \frac{e}{R}$

Solution :

By using $i = \frac{e}{R} = -\frac{1}{R}\frac{d\phi}{dt}$

$i = -\frac{1}{10}\frac{d}{dt}\left(6t^2 - 5t + 1\right) = -\frac{1}{10}(12t - 5)$

$i = -\frac{1}{10}(12 \times 0.25 - 5) = 0.2A$

Final Answer : (4)

30) **Question Explanation :**

Given the i - t graph for a coil, we have to determine the graph for the induced emf with time.

Concept: Self Inductance (Induced emf vs time Graph)

Solution:

$\mathcal{E} = -L\frac{di}{dt}$ =-(L)(Slope of i-t graph)

Where L is the inductance and $(\frac{di}{dt})$. is the rate of change of current.

Lets analyze slope of I-t graph:

Region 1 (horizontal line):

Current constant $\Rightarrow \frac{dI}{dt} = 0$

$\Rightarrow emf = 0$

Region 2 (linearly decreasing):

Current decreasing ⇒ slope negative

$\Rightarrow \frac{dI}{dt} < 0$

$\Rightarrow e = -L\frac{dI}{dt} \Rightarrow$ positive constant emf

Region 3 (linearly increasing):

Current increasing ⇒ slope positive

$\Rightarrow \frac{dI}{dt} > 0$

$\Rightarrow$ emf = negative constant

So emf graph will be:

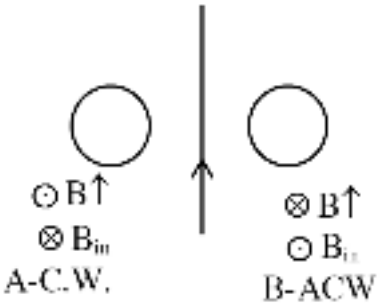
First → zero

Then → positive constant (rectangle above axis)

Then → negative constant (rectangle below axis)

Final Answer: Option (3)

31)



32) **Question Explanation:** A solenoid carries AC current. Outside the solenoid, changing magnetic flux induces an electric field. Find Its magnitude at distance r > R.

Concept: From Faraday's law: induced electric field around a closed loop outside solenoid is due to changing flux inside the solenoid.

Solution:

$$\oint \mathbf{E}\cdot d\mathbf{l} = -\frac{d\Phi_H}{dt} \quad \rightarrow \quad E\cdot 2\pi r = -\pi R^2\mu_0 n\frac{dI}{dt}$$

$$E = \frac{\mu_0 n R^2}{2r}\frac{dI}{dt}$$

$$\text{Given } I = I_0\cos\omega t \rightarrow \frac{dI}{dt} = -I_0\omega\sin\omega t.$$

Solution:

$$E = \frac{\mu_0 n R^2}{2r}\cdot I_0\omega\sin\omega t$$

Final Answer: Option (1)

33) As the magnet falls, due to the variation of magnetic flux, induced current is developed in the hollow conductor so as to resist the cause of increment of flux; that means the velocity of the magnet is resisted. In other words we can say that the magnet is retarded before entering into the hollow conducting cylinder. Consequently its acceleration will be less than g because magnetic force opposes the gravitational force acting on the magnet. If the cylinder is too long, the magnetic force opposes the gravitational force on the magnet so as the yield zero acceleration after a long time to attain the terminal velocity. The eddy current in the cylinder results in its heating.

34)

$$\therefore L = \frac{BNA}{I} = \frac{\mu_0 Ni}{2R}\times\frac{A}{I}$$

$$L = \frac{\mu_0 N\pi R^2}{2R} \Rightarrow L\propto R$$

$$\therefore L\propto\sqrt{A}$$

35) **Question Explanation :** Three inductors having a value of 3.0 H, are connected in parallel. We need to find the equivalent inductance of the parallel combination.

Concept: Combination of inductors (Series and parallel)

Solution:

For inductors in parallel, the equivalent inductance is given by $\Rightarrow \frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$
If all three inductors are 3.0 H each, $L_1 = L_2 = L_3 = 3.0$ H, the calculation is:
$\frac{1}{L_{eq}} = \frac{1}{3.0} + \frac{1}{3.0} + \frac{1}{3.0} = \frac{3}{3.0}$
$L_{eq} = 1H$

Final Answer: option (1).

36) **Question Explanation:** In this question, we are asked to find the self-inductance of a toroid in terms of its dimensions and number of turns.

Concept: Self-Induction And Self-Inductance Of Coil/Solenoid/Toroid

Solution :
Magnetic field Inside a Toroid (B) :
Let N be the total number of turns and R_m be the mean radius of the toroid.
$B = \frac{\mu_0 NI}{2\pi R_m}$

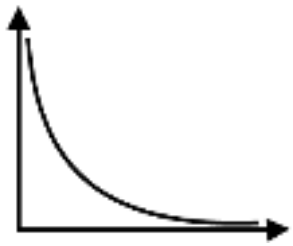
Magnetic Flux Through a Single Turn (ϕ):
If r is the radius of the cross-section of the toroid, the cross-sectional area is $A = \pi r^2$.
$\phi = B \cdot A = \left(\frac{\mu_0 NI}{2\pi R_m}\right) \cdot \left(\pi r^2\right)$
$\phi = \frac{\mu_0 NIr^2}{2R_m}$

Total Flux Linkage (Φ_{total}) :
$\Phi_{total} = N\phi = N\left(\frac{\mu_0 NIr^2}{2R_m}\right) = \frac{\mu_0 N^2 Ir^2}{2R_m}$

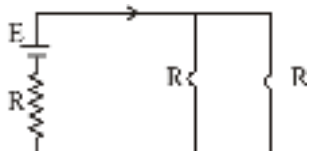
Self-Inductance (L) :
By definition of self-inductance, $\Phi_{total} = LI$:
$L = \frac{\Phi_{total}}{I} = \frac{\mu_0 N^2 r^2}{2R_m}$

Final Answer: Option (1)

37) $emf = \frac{-Ldi}{dt}$



38)



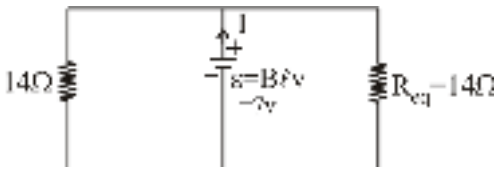
$E = B\ell V$

$I = \frac{\varepsilon}{R_{eq}}$

$I = \frac{\varepsilon}{3R/2} = \frac{2\varepsilon}{3R}$

$I = \frac{2B\ell V}{3R}$, $I_1 = I_2 = \frac{I}{2} = \frac{B\ell V}{3R}$

39)



$\varepsilon = B\ell v = 1 \times 7 \times 1 = 7v$

$R_{total} = \frac{14}{2} = 7\Omega$

Current from battery

$I = \frac{\varepsilon}{R_{total}} = \frac{7}{7}A$

$I = 1A$

40)

For transformer $\frac{N_S}{N_P} = \frac{E_S}{E_P}$

$\Rightarrow E_S = E_P \times \frac{N_S}{N_P}$

$\Rightarrow E_S = 80 \times \frac{5000}{500}$

$E_S = 800$ V

and frequency remain 50 Hz

41) Effective length of the wire is 2L

$F = i B (2L)$

42) **Question Explanation:** Find orientation of electric and magnetic fields so electron passes undeflected.

Concept: For undeflected motion, electric and magnetic forces must be equal and opposite.

Formula
qE = quB
$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$
Solution:
Electron moves right.
Need electric and magnetic forces opposite.
$\vec{B}$ into page
$\vec{v}$ right
Then
$\vec{v} \times \vec{B}$ = upward for positive charge
For electron:
$\vec{F}_B$ = downward
Electric field downward gives force on electron upward.
Thus forces oppose each other.

Final Answer : Option (3)

43) **1. Question Explained :** A wire bent into a loop then its magnetic moment is.
2. Concept :
This concept based on Magnetic moment.

3. Formula Used :
M = iA

4. Solution:
From = 2πR
we get R = $\frac{\ell}{2\pi}$
Area = πR^2
= $\pi \frac{L^2}{4\pi^2}$
Area = $\frac{L^2}{4\pi}$
$M = \frac{IL^2}{4\pi}$

Hence,
Correct option: (1)

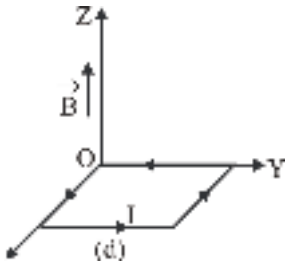
44)

Explain the Question : A rectangular loop of current I is placed in a uniform magnetic field B along the +z direction. Different orientations of the loop are shown. We must identify which orientation corresponds to stable equilibrium.

Concept : This question is based on torque and equilibrium of a magnetic dipole in uniform

magnetic field.

Solution :



Torque on the loop in magnetic field $\vec{\tau} = \vec{M} \times \vec{B}$.
For equilibrium, $\vec{\tau} = 0 \Rightarrow \vec{M}$ parallel or antiparallel to $\vec{B}$.
Stable equilibrium occurs when magnetic moment is parallel to the magnetic field (minimum potential energy).
Examining the given cases:
Only in case (d), the normal to the loop (and hence magnetic moment) is in the direction of +z axis.
All other cases correspond to either unstable equilibrium or non-equilibrium.

Final answer : Option (4)

45) **Explanation the Question:**
Proton has velocity with a component parallel to the field (along Y) and a perpendicular component; perpendicular part makes uniform circular motion, so overall path is a helix.

Concept: This question is based on Radius form $r = \frac{mv_\perp}{qB}$. Period from T = $\frac{2\pi m}{qB}$. Parallel velocity gives axial advance.
Formula:
$v_\perp = v \cos 60 = 1.0 \times 10^6$ m/s (perpendicular to B)
$v_{||} = v \sin 60 = 1.732 \times 10^6$ m/s (parallel to B)
$r = \frac{mv_\perp}{qB}$
$T = \frac{2\pi m}{qB}$

Solution:
$r = \frac{1.67 \times 10^{-27} \times 1.0 \times 10^6}{1.6 \times 10^{-19} \times 0.104} \approx 0.10m$
$T = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.104} \approx 2\pi \times 10^{-7}s.$
Final Answer : Option (3)

CHEMISTRY

46) $\frac{\pi = iCRT}{\Delta T_f = KiC} = \pi = \frac{RT \times \Delta T_f}{K}$

$$= \frac{0.082 \times 300 \times 0.4}{1.8} = 5.55$$

47) $\pi \propto M \quad \pi_{final} = \frac{\pi_1 V_1 + \pi_2 V_2}{V_1 + V_2} = \frac{1.64v + 2.46v}{2v}$
$\pi_{final} = 2.05$ atm

48) In negative deviation solution, intermolecular force after mixing (A – B) is greater than inter molecular force before mixing (A – A, B – B)

49) NCERT BASED

50)

$\Delta T_b = iK_bm$
• Consider $m \approx M$
• K_b same for all
$i \times M$
(1) $1 \times 0.01 = 0.01$
(2) $2 \times 0.01 = 0.02$
(3) $3 \times 0.01 = 0.03$
(4) $1 \times 0.015 = 0.015$
$iM \uparrow \Rightarrow \Delta T_b \uparrow \Rightarrow T_b \uparrow$

51) Van't Hoff factor 'i' for urea = 1 and for NaCl = 2
$\pi = iCRT$
Urea $i \times M = 1 \times 0.1 = 0.1$
NaCl $i \times M = 2 \times 0.1 = 0.2$
$iM \uparrow \pi \uparrow$
O.P. is a C.P.

52)

A. Question Explanation:

The question asks us to identify the pair of solutions that are isotonic. Isotonic solutions have the same osmotic pressure. For dilute solutions, osmotic pressure is proportional to the total concentration of dissolved particles (ions or molecules).

B. Given Data:

A. A: 0.1 M glucose (glucose is a non-electrolyte, so it doesn't dissociate)

B. B: 0.05 M KCl (KCl dissociates into 2 ions: K^+ and Cl^-)

C. C: 0.05 M $MgCl_2$ ($MgCl_2$ dissociates into 3 ions: Mg^{2+} and $2Cl^-$)

D. D: 0.1 M $AlCl_3$ ($AlCl_3$ dissociates into 4 ions: Al^{3+} and $3Cl^-$)

C. **Concept**: Isotonic solutions, osmotic presser.

D. Mathematical Calculation:

We need to calculate the effective molarity of particles for each solution:

A. A (glucose): 0.1 M * 1 particle/molecule = 0.1 M

B. B (KCl): 0.05 M * 2 ions/molecule = 0.1 M

C. C ($MgCl_2$): 0.05 M * 3 ions/molecule = 0.15 M

D. D ($AlCl_3$): 0.1 M * 4 ions/molecule = 0.4 M

E. Final Answer:

Solutions A (0.1 M glucose) and B (0.05 M KCl) both have an effective particle concentration of 0.1 M. Therefore, they are isotonic. The answer is option 1.

53)

1. Understand the Concept:

Depression in freezing point (ΔT_f).

Solution:

$\Delta T_f = K_f \times m$

2. Calculate ΔT_f :

ΔT_f = Freezing point of pure nitrobenzene – Freezing point of the solution ΔT_f = 278.82 K – 276.82 K ΔT_f = 2.0 K

3. Rearrange the Formula to Solve for K_f :

$K_f = \Delta T_f / m$

4. Plug in the Values and Calculate:

K_f = 2.0 K / 0.25 mol/kg K_f = 8.0 K kg/mol

54) $\Delta T_{f_1} = \Delta T_{f_2}$

$m_1 = m_2$

$$\frac{0.6 \times 1000}{60 \times 100} = \frac{1.2 \times 1000}{M_{w_M} \times 100}$$
$$M_{w_M} = 120$$

55) RLVP = X_B

56)

Question Explanation: Compare solubility using Henry's law constants.

Concept:
Solubility ∝ 1 / K_H
Lower K_H → higher solubility

Solution:
K_H values: B (2×10^{-5}) < C (35) < A (145)

So solubility: B > C > A

Final Answer: Option (2)

57) $$M = \frac{\%\frac{w}{w} \times d \times 10}{M_{w,solute}} = \frac{36.5 \times 1.2 \times 10}{36.5}$$
M = 12 mol/L

58) **Correct Statements:**

A. **(i) A catalyst does not alter the equilibrium state.**
   A. A catalyst speeds up both the forward and reverse reactions equally, so it doesn't affect the position of equilibrium. It only helps the system reach equilibrium faster.
B. **(iii) For any reaction greater the value of equilibrium constant greater is the extent of completion of reaction.**
   A. A larger equilibrium constant (K) indicates that the reaction favors the formation of products at equilibrium. This means a higher proportion of reactants will be converted into products when the reaction reaches equilibrium.
C. **(iv) For the reaction, N204 ⇌ 2N02, Kc = Kp/RT.**
   A. The correct relationship between Kp and Kc is: Kp = Kc x $(RT)^{\Delta n}$ where Δn is the change in the number of moles of gas during the reaction. For the given reaction, Δn = 1, so Kp = Kc x RT.
D. **(vi)The value of equilibrium constant does not depend upon pressure.**
   A. This statement is true for reactions. Equilibrium constant depend only temperature .

Incorrect Statements:

A. **(ii) For the equilibrium, $H_2 + I_2 \rightleftharpoons 2HI$, the value of equilibrium constant increases with increase in concentration of H_2.**
   A. The equilibrium constant (K) is a constant at a specific temperature and is independent of the initial concentrations of reactants.
B. **(v) For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the degree of dissociation of PCl_5 increases with**

increase in pressure.

A. This reaction involves an increase in the number of moles of gas. According to Le Chatelier's principle, increasing the pressure will shift the equilibrium towards the side with fewer[1] moles of gas to minimize the effect of the pressure increase. Therefore, the degree of dissociation of PCI_5 will decrease with an increase in pressure.

Therefore, only 4 statements are correct.

59) Shifting of equilibrium will depends upon stoichiometry of reactants and products.
At constant pressure, introduction of inert gas causes volume increase due to which partal pressure of various reactant and product decreases.

60) **Explanation:**

$PCl_{5\,(g)} \rightleftharpoons PCl_{3\,(g)} + Cl_{2\,(g)}$

Initially 1 mol

at equilibrium: 1 - x x x

$\alpha = 80\%$ or 0.8

1 - 0.8=0.2 0.8 0.8

$$Kc = \frac{(PCl_3)(Cl_2)}{(PCl_5)} = \frac{(0.8)(0.8)}{(0.2)}$$

$$\boxed{K_c = 3.2}$$

Final answer : 3.2

61) $AB_3(g) \rightleftharpoons AB_2(g) + \frac{1}{2} B_2(g)$

t = 0 800 0 0

Eqn. 800 – x x $\frac{x}{2}$

$$P_{Total} = 800 - x + x + \frac{x}{2} = 900$$

$$\frac{x}{2} = 100$$

x = 200 torr

$$\% \text{ of } AB_3 \text{ dissociated} = \frac{200}{800} \times 100 = 25\%$$

62) $2A \rightleftharpoons B + C$; $K_c = 4 \times 10^{-3}$

At a given time ⇒ [A] = [B] = [C] = 2×10^{-3} M

$$Q_c = \frac{[B][C]}{[A]^2}$$

$Q_c = 1$

as $Q_c > K_c$

Reaction has a tendency to go in backward direction.

63) $K_p = K_c(RT)^{\Delta ng}$

when $\Delta n_g = 0$ then $K_p = K_c$

$\Delta n_g \neq 0$ $K_p \neq K_c$

(1) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

$\Delta n_g \neq 0$

$K_p \neq K_c$

64)

Explain the question
Analyze ice-water equilibrium shifts.

Concept
Le Chatelier's Principle.

Solution
For the equilibrium $H_2O_{(s)} \rightleftharpoons H_2O_{(l)}$, the forward process (melting) is endothermic ($\Delta H>0$); thus, increasing temperature shifts the equilibrium forward. Regarding pressure, water exhibits anomalous behavior where the density of ice is less than liquid water, meaning the volume of ice is greater than the volume of water ($V_{ice} > V_{water}$). According to Le Chatelier's Principle, increasing pressure shifts the equilibrium toward the side with lower volume (the liquid phase). While the Assertion is true because both factors favor water formation, the Reason is false because it incorrectly claims the volume of water is greater than the volume of ice.

Final answer : Option 2

65)

Question Explanation:
The question asks to identify the incorrect statement about chemical equilibrium from the given options.

Concept :
This question is based on Measurable properties it equilibrium .

Solution :

A. 1. On addition of catalyst the equilibrium constant value is not affected:

Correct: Catalysts affect the rate of reaction, not the position of equilibrium.

B. 2. Equilibrium constant for a reaction with negative ΔH value decreases as the temperature increases:

A. Correct: A negative ΔH indicates an exothermic reaction. Increasing temperature favors the reverse (endothermic) reaction, decreasing the equilibrium constant.

C. 3. In equilibrium mixture of ice and water kept in perfectly insulated flask, mass of

ice and water does not change with time:

A. Correct: At equilibrium, the rates of melting and freezing are equal, so the masses remain constant. In an insulated flask, there's no heat exchange with the surroundings.

D. 4. All the measurable properties of the system are variable at equilibrium state:

Incorrect: At equilibrium, all measurable properties are constant. They do not change with time.

Conclusion:

The statement that is not true is "All the measurable properties of the system are variable at equilibrium state."

Hence,
Option (4) is correct

66) With change of pressure, x will change in such a way that K_P remains constant because K_p depends only on temperature.

67) **Question Asking About:**
Find minimum solubility in all four options.

Concept:
Minimum solubility of AgBr occurs when a common ion is added to the solution.

Solution/Explanation: Option (2): $S = \frac{K_{SP}}{2C}$
Option (3): $S = \frac{K_{SP}}{C}$
Option (4): $S = \frac{K_{SP}}{C}$ 0.1 M Ca Br_2

68) **Solution/Explanation:**
$FeCl_3 + H_2O \rightarrow Fe(OH)_3 + 3HCl$ **acidic**
$NH_4Cl + HOH \rightarrow NH_4OH + HCl$ **acidic**
$\overset{+}{N}H_4CH_3COO^- \rightarrow CH_3COOH + NH_4OH$ almost neutral
$CO_3^{-2} + 2H_2O \rightarrow H_2CO_3 + 2OH^-$ Basic

69) Old NCERT Pg. # 212

70) Solution/Explanation/Calculation
Explanation:

Find P^H of indicator

Concept: $$P^H = P^K In + \log \frac{[In^-]}{[HIn]}$$

$$-\log \frac{[In^-]}{[HIn]} = P^K In - P^H$$

Final Answer: $$\log \frac{[HIn]}{[In^-]} = P^K In - P^H$$

71)

$H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-$; Ka_1

$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$; Ka_2

$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}$; Ka_3

_____

$H_3PO_4 \rightleftharpoons 3H^+ + PO_4^{3-}$; $K = Ka_1 \times Ka_2 \times Ka_3$

For polybasic acid, order of ionization constant is $Ka_1 > Ka_2 > Ka_3$ because as negative charge increases, tendency to loose H^+ ions decreases.

Overall $K = Ka_1 \times Ka_2 \times Ka_3$

Taking log on both sides

$\log K = \log Ka_1 + \log Ka_2 + \log Ka_3$

72)

Question Explanation : We need to find molar solubility of nickel hydroxide in presence of common hydroxide ions using K_{sp} value

Concept : Common ion effect Based solubility

Solution :

$Ni(OH)_2 \rightleftharpoons Ni^{+2} + 2OH^-$

s 2s

$K_{sp} = [Ni^{+2}][OH^-]^2 = (s)\ (2s)^2$

Molar solubility = s

NaOH provides 0.10 M OH, which is much larger than 2s, so $[OH^-] = 0.10 + 2s \approx 0.10$ M

$K_{sp} = (s)\ (0.10)^2$

$$s = \frac{K_{sp}}{(0.10)^2} = \frac{2 \times 10^{-15}}{(0.10)^2} = 2 \times 10^{-13}$$

$s = 2 \times 10^{-13}$ M

Final answer : Option 1

73) **Explanation :** pH is to be found of mixture of two acidic solution.

Given data : $(pH) = 3$, $V_1 = 100$ mL

$(pH)_2 = 3$, $V_2 = 400$ mL

Concept : $pH = -\log (H^+)_{mix}$.

For (Acid + Acid) ⇒ $$(H^+) = \frac{N_1V_1 + N_2V_2}{V_1 + V_2}$$

Calculation :

$(H^+)_1 = 10^{-3}$, $V_1 = 0.1$ L
$(H^+)_2 = 10^{-3}$, $V_2 = 0.4$ L

$$(H^+) = \frac{10^{-4} + 4 \times 10^{-4}}{0.5} = 10^{-3}$$

$$pH = -\log\left(10^{-3}\right) = 3$$

Final Answer : 3.

74)

Explain the Question : Need to find degree of hydrolysis is independent from concentration.
Concept : This question is based on hydrolysis of WAWB slat.
Solution : The 'h' for salt of WAWB type is independent from concentration.

$$h = \sqrt{\frac{K_w}{K_a \times K_b}}$$

CH_3COONH_4
Final answer : 3

75)

Question Explanation: pH of the solution formed after mixing strong acids (H_2SO_4, HCl) with strong base $Ba(OH)_2$ is to be calculated using gram-equivalent concept.

Concept:
Neutralization occurs on equivalent basis.
Excess gram-equivalents of H^+ or OH^- decide the pH.

Solution:
H_2SO_4 (M/200), 10 mL → Normality = M/100
Acid eq = (1/100) × 10 = 0.10 mmol H^+

HCl (M/100), 40 mL → Normality = M/100
Acid eq = (1/100) × 40 = 0.40 mmol H^+

Total acid eq = 0.50 mmol H^+

$Ba(OH)_2$ (0.004 M), 50 mL → Normality = 0.008 N
Base eq = 0.008 × 50 = 0.40 mmol OH^-

Excess H^+ eq = 0.50 − 0.40 = 0.10 mmol

Total volume = 100 mL
$[H^+]$ = 0.10 / 100 = 0.001 M
pH = 3

Final Answer: Option 2

76)

Explain the Question : We need to find pH of the solution.

Concept : This question is based on buffer solution.

Solution : $pH = pKa + \log \frac{[KCN]}{[HCN]}$

Since volume is same for both

$pH = 9.3 + \log\left(\frac{2.5}{2.5}\right)$

pH = 9.3

Final answer : 1

77) $pOH = pK_b + \log \frac{[C.A]}{[W.B]} = 4.76 + \log \frac{0.1}{0.2} = 4.46$

pH = 14 – 4.46 = 9.54

78)

Pka = 5 – log2 = 4 – 7

CH_3COONa = WASB salt

$pH = 7 + \frac{1}{2}(pK_a + \log C)$

$= 7 + \frac{1}{2}(4.7 + \log 10^{-2})$

pH = 8.35

79)

Explanation

The pH of a solution is a measure of the hydrogen ion ($[H^+]$) concentration and is given by the formula: $pH=-\log[H^+]$

A decrease in pH means an increase in the $[H^+]$ concentration.

Given Data

• Initial pH = 6

• Dilution factor = 100 times

Concept

• pH formula: $pH=-\log[H^+]$

• Effect of dilution on pH: When a solution is diluted 100 times, the concentration of H^+ ions decreases by a factor of 100.

v Contribution of water auto-ionization:

Water itself contributes $[H^+]=10^{-7}$M, because $K_w=[H^+][OH^-]=10^{-14}$

Calculation

1. Find the initial $[H^+]$ concentration:$[H^+]=10^{-6}$ M

2. New concentration after dilution (100 times):$[H^+]=10^{-6}/100=10^{-8}$ M

3. Total $[H^+]$:

$[H+]_{total}=10^{-8}+10^{-7}=1.1\times10^{-7}$M

4. New pH : $pH=-\log(1.1\times10^{-7})$ = 7-Log(1.1) =6.98

Answer option 1, (6.98)

80)

Question Explanation: We need to find pH when 0.4 g of NaOH is dissolved in 10 L solution.

Concept: pH calculation of strong base solution.

Solution:
NaOH is a strong base → completely ionizes.
pOH = $-\log[OH^-]$, pH + pOH = 14

Molar mass of NaOH = 40 g mol^{-1}

Moles of NaOH = 0.4 / 40 = 0.01 mol

Volume = 10 L

Molarity = Moles/Volume(L) = 0.01 / 10 = 0.001 M = 10^{-3} M

[NaOH] = $[OH^-] = 10^{-3}$ M

pOH = $-\log(10^{-3}) = 3$

pH = 12 − 3 = 9 (T = 90°C)

Final Answer: Option (4)

81)

Explain Question:
Determining the effect of heating on the pH of pure water.

Concept:
Auto-ionization of water and Le Chatelier's Principle.

Solution:
• The dissociation of water ($H_2O \rightleftharpoons H^+ + OH^-$) is an endothermic process.
• According to Le Chatelier's Principle, increasing the temperature shifts the equilibrium in the forward direction.
• This increases the concentration of both $[H^+]$ and $[OH^-]$. ions.
• Since pH = –log$[H^+]$, an increase in $[H^+]$ results in a decrease in the pH value.
• Note: Water remains neutral because $[H^+]$ still equals $[OH^-]$., but the neutral point shifts below 7.

Final answer: Option(1)

82)

<u>Explanation</u>
The **pH** of a solution is determined by the concentration of hydrogen ions ($\mathbf{H^+}$) it releases.
1. Hydrochloric acid (HCl) is a **strong acid**, meaning it completely ionizes in water:
$HCl \rightarrow H^+ + Cl^-$
This results in a high $\mathbf{H^+}$ concentration and thus a **low pH**.
2. Acetic acid ($\mathbf{CH_3COOH}$) is a **weak acid**, meaning it only partially ionizes in water:

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

Since it does not fully ionize, the **H^+ concentration is lower**, leading to a **higher pH** compared to HCl of the same concentration.

• **Assertion:** *pH of hydrochloric acid solution is less than that of acetic acid solution of the same concentration.* (True)

• Since HCl completely ionizes and acetic acid only partially ionizes, HCl has a higher H^+ concentration, making its pH lower.

• **Reason:** *In equimolar solutions, the number of titrable protons present in hydrochloric acid is less than that present in acetic acid.* (False)

• Acetic acid only **partially ionizes**, meaning **fewer** free protons are available initially.

Thus, the **number of titrable protons in HCl is greater, (not lesser) than in acetic acid**.

Answer option 3

83)

A. **Question Explanation**:

We need to determine which of the given mixtures forms a buffer solution. A buffer solution resists changes in pH when small amounts of acid or base are added. It is typically composed of:

A. A weak acid and its conjugate base (e.g., CH_3COOH + CH_3COO^- from CH_3COONa).
B. A weak base and its conjugate acid (e.g., NH_3 + NH_4^+).

B. **Given Data**:

A. Option 1: 100 ml, 0.1N CH_3COOH + 50 ml, 0.1N NaOH
B. Option 2: 100 ml, 0.1N HCl + 100 ml, 0.1N NaOH
C. Option 3: 50 ml, 0.1N CH_3COOH + 100 ml, 0.1N NaOH
D. Option 4: 100 ml, 0.1N HCl + 50 ml, 0.1N NaOH

C. **Concept**:

A. Buffer Formation Condition:
    A. A buffer forms when a weak acid (like CH_3COOH) is partially neutralized by a strong base (like NaOH), producing its conjugate base (CH_3COO^-).
    B. A strong acid (HCl) and a strong base (NaOH) completely neutralize each other and do not form a buffer.
B. Analysis of Options:
    A. Option 1: CH_3COOH is a weak acid. It reacts with NaOH partially (since NaOH moles < CH_3COOH moles), forming CH_3COO^-, creating a buffer.
    B. Option 2: HCl and NaOH are both strong, leading to complete neutralization (no buffer).
    C. Option 3: Excess NaOH fully neutralizes CH_3COOH and leaves extra OH^-, meaning no buffer is formed.
    D. Option 4: HCl is in excess, so no weak acid-conjugate base pair is formed.

D. **Mathematical Calculation**:

A. Moles of CH_3COOH in Option 1:

   A. 100 ml × 0.1 N = 10 milliequivalents

A. Moles of NaOH in Option 1:

   A. 50 ml × 0.1 N = 5 milliequivalents

A. Reaction: $CH_3COOH + NaOH \rightarrow CH_3COO^- + H_2O$

   A. 5 milliequivalents of NaOH neutralize 5 milliequivalents of CH_3COOH.
   B. Remaining CH_3COOH = 5 milliequivalents, and CH_3COO^- formed = 5 milliequivalents.
   C. This forms a buffer solution (weak acid + conjugate base).

E. **Final Answer**:
Option 1: 100 ml, 0.1N CH_3COOH + 50 ml, 0.1N NaOH

F. **Question Level**:
EASY

84)

Explanation:
The question asks about the change in pH value when sodium acetate is added to an aqueous solution of acetic acid.

Concept:
Common Ion Effect: The suppression of the ionization of weak electrolyte caused by the addition of common ion from a strong electrolyte.
$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$
$CH_3COONa \rightarrow CH_3COO^- + Na^+$
The addition of sodium acetate increases the concentration of acetate ions (CH_3COO^-) in the solution. This increase in acetate ion concentration shifts the equilibrium of the acetic acid ionization reaction to the left (Le Chatelier's principle).
Decrease in H^+ Ion Concentration: As the equilibrium shifts to the left, the concentration of H^+ ions decreases.
Increase in pH Value: Since pH is inversely related to the H^+ ion concentration, a decrease in H^+ ion concentration results in an increase in pH value.
Final Answer: pH value increases. Correct Option:4

85)

Explain Question:
Identifying the solution with the highest pH (most basic) among given salts.

Concept:
Salt Hydrolysis and Acid-Base nature of salts.

Solution:

• HCOOK (Potassium Formate): Salt of a weak acid (HCOOH) and a strong base (KOH).
It undergoes anionic hydrolysis to form a basic solution (pH > 7)
• MgCl2 (Magnesium Chloride): Salt of a strong acid (HCI) and a relatively weak base (Mg(OH)2).
It is slightly acidic (pH < 7)
• NaNO3 (Sodium Nitrate): Salt of a strong acid (HNO3) and a strong base (NaOH).
It is neutral (pH = 7)
• FeSO4 (Ferrous Sulfate): Salt of a strong acid (H2SO4) and a weak base (Fe(OH)2).
It is acidic (pH < 7)
Since HCOOK is the only basic salt, it will have the maximum pH.

Final answer: Option(1)

86)

Explain the Question : Need to find dissociation constant (K_a)
Concept : This question is based on ostwald's dilution law.
Solution : $K_a = C \propto^2$
$K_a = 0.02 \times (0.01)^2$
$= 2 \times 10^{-6}$
Final answer : 1

87)

Explain Question : Find the pH of a 0.1 M NaI solution.
Concept : This question is based on NaI is a neutral salt of strong acid (HI) and strong base (NaOH); pH ≈ 7.
Solution :
Since NaI fully dissociates and neither Na^+ nor I^- hydrolyzes, solution is neutral.
Final answer : 4

88)

Explain Question: Calculating the degree of dissociation of a weak base using pH and concentration.

Concept: Ionization of Weak Bases and pOH.

Solution:
1. Calculate pOH:
a. pH + pOH = 14
b. pOH = 14 – pH = 14 – 11 = 3
2. Calculate $[OH^-]$ concentration:
a. $[OH^-] = 10^{-pOH} = 10^{-3}$ M
3. Calculate the degree of dissociation (α):
a. $\alpha = [OH^-] / [NH_3] = 10^{-3} / 0.05 = 0.02$
4. Convert a to percentage:
a. Percentage dissociation = α × 100% = 0.02 x 100% = 2%

Final answer: Option (1)

89)

Question Explanation: Effect of NH_3 on solubility of AgBr.

Concept: Solubility due to complex formation.

Solution:
AgBr is sparingly soluble.
In presence of NH_3, Ag^+ forms complex ion:
$AgBr(s) \rightleftharpoons Ag^+ + Br^-$
$Ag^+ + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+$
Formation of complex ion decreases free Ag^+ concentration, so equilibrium shifts forward and solubility increases.
Statement-I: Solubility of AgBr decreases in presence of NH_3 (Incorrect, it increases)
Statement-II: AgBr forms complex $[Ag(NH_3)_2]^+$ in presence of NH_3 (Correct)
Statement-I is incorrect but statement-II is correct.

Final Answer: Option (3)

90) **Explain Question:**
Question is asking about preparation of acidic buffer.

Concept:
P^H of acidic buffer solution.

Solution:
Acidic buffer is prepared by mixing weak acid and its salt with strong base.
weak acid and its conjugate base

Final answer: Option (1)

BIOLOGY

91) NCERT (XII) Page# 115/127(H)

92) NCERT (XII) Page# 113/124(H)

93) NCERT (XII) Page# 115/126(H)

94) NCERT (XII) Page# 111/122-123(H)

95) NCERT (XII) Page# 111/126(H)

96) NCERT (XII) Page# 110/121(H)

97) NCERT (XII) Page# 111/122(H)

98) NCERT (XII) Page# 111/122(H)

99) NCERT (XII) Page# 111/122(H)

100) NCERT (XII) Page# 111-112/123(H)

101) NCERT (XII) Page# 113/124(H)

102) NCERT (XII) Page# 113/124(H)

103) NCERT (XII) Page# 115/126(H)

104) NCERT (XII) Page# 115/126(H)

105) NCERT (XII) Page# 115/127(H)

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108) NCERT (XII) Page# 121/133(H)

109) NCERT (XII) Page# 122/133(H)

110) NCERT (XII) Page# 125/136(H)

111)

Module-7 Pg#212

112) NCERT (XIIth) Pg. # 133

113) NCERT (XIIth) Pg. # 137 / Para. # 7.7

114) NCERT XII, Pg.# 134

115) NCERT XII, Pg.# 135

116) NCERT(XII) Pg.#138, Ist para

117) NCERT(XII) Pg # 137/148(H) Para:7.7

118) NCERT(XII) Pg#140/151(H) Para:7.9

119)

NCERT XII, Page # 128

120) NCERT XII Pg # 130

121) **Question Explanation**: Evaluate whether the survival of pouched mammals in Australia is due to absence of competition.
Concept: The question is based on **adaptive radiation**
Solution

A. **Assertion**: Pouched mammals survived in Australia — True
B. **Reason**: Lack of competition from other mammals — True
C. Since placental mammals didn't reach Australia due to geographic isolation, marsupials diversified without competition.
D. The **reason correctly explains** the assertion.

Final Answer: Option 1

122) **Question Explanation:** Meteorite analysis and chemical evolution.
Concept: Extraterrestrial origin of life.
Solution: The **theory of chemical evolution** suggests that organic molecules, the building blocks of life, could have formed under primitive Earth conditions. The discovery of similar organic compounds on meteorites, like the **Murchison meteorite**, provides evidence that the processes leading to the formation of these molecules are not unique to Earth but can occur elsewhere in the universe. This supports the idea that life's building blocks could have also come from space.
Final Answer: Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

123) **Question Explanation**: Evaluate whether the statements correctly describe homology and its evolutionary significance.
Concept: The question is based on homologous structures and common ancestry.

Solution:

A. Statement-I is correct because thorns of Bougainvillea and tendrils of Cucurbita are modified stems showing structural similarity.
B. Statement-II is incorrect because homology does indicate common ancestry.

Final Answer: Option 2

124) NCERT Pg # 124

125) NCERT XII Page No. # 118

126) NCERT XII Page No. # 122

127) NCERT (XII) Bio Pg # 131

128) NCERT Pg # 133, para 7.4, fig. 7.6

129)

Question Explanation : Identify lineages in the provided phylogenetic tree.
Concept : Brief Account Of Evolution.
Solution :
• A is the earliest branch (ancestor to mammals) → Synapsids.
• B (from synapsid line, before mammals) → Pelycosaurs.
• C (from the sauropsid line, leading towards dinosaurs/birds) → Thecodonts.
Thus, the correct identification is : A = Synapsids, B = Pelycosaurs, C = Thecodonts.
Final Answer : Option (4) The correct identification is A-synapsids, B-pelycosaurs, and C-thecodonts.

130)

A. Query successful

Question Explanation: Matching major events to their timeline.**Concept:** Timeline of Evolution
Solution: Here's the correct timeline matching:

A. (A) The **Origin of the Universe** (Big Bang) occurred approximately **20 billion years ago** (iv).
B. (B) The **Origin of Earth** occurred about **4.5 billion years ago** (ii).
C. (C) The first **cellular forms of life** appeared around **2 billion years ago** (v).
D. (D) The first **non-cellular forms of life** (e.g., RNA, proteins) appeared around **3 billion years ago** (iii).
E. (E) **Life appeared** about **4 billion years ago** (i), as non-cellular forms.

Final Answer: A - iv, B - ii, C - v, D - iii, E - i

131) NCERT (XII) Pg. # 127, 128

132)

Explain the Question : The question checks the correctness of statements related to age of the universe, Earth, evolution, and spontaneous generation.
Concept : This question is based on origin and evolution of life.
Solution :
(A) Evolutionary biology studies history of life → True
(B) Universe is ~13.8 billion years old → True
(C) Earth formed ~4.5 billion years ago, not 4.0 → False
(D) Spontaneous generation explains origin of life, not universe → False
Final answer : 3

133)

NCERT XII Pg. # 124-125

134) **Question Explanation :** Count the number of mammalian pairs showing convergent evolution. **Concept :** The question is based on convergent evolution (analogous adaptations in unrelated groups).
Solution : The correct answer is option 2.
• Mouse – Marsupial mouse → shows convergent evolution (placental vs marsupial).
• Numbat – Spotted cuscus → does not show convergent evolution.
• Mole – Marsupial mole → shows convergent evolution (burrowing adaptation).
• Lemur – Anteater → does not show convergent evolution.
• Flying squirrel – Flying phalanger → shows convergent evolution (gliding adaptation).
• Bob cat – Koala → does not show convergent evolution.
Total pairs showing convergent evolution = 3.
Final answer : option 2

135)

Question Explanation: The trend of increasing brain size in human evolution.

Concept: Human Evolution.

Solution: There has been a general trend of increasing cranial capacity during human evolution. Homo habilis had a brain size of about 650-800 cc, followed by Homo erectus at about 900 cc, and then Homo sapiens with an average of 1350-1400 cc. The correct order is a, b, c.

Final Answer: a, b, c.

136) **Explain Question :** Metagenesis means
Concept : This question is based on metagenesis
Explanation : In Obelia and other cnidarians, polyps reproduce asexually to form medusae, while medusae reproduce sexually to form polyps.

Final answer : 1: Polyp produce medusae asexually and medusae produce polyp sexually

137)

Question Explanation: Identifying the animal with a muscular pharynx.
Concept: Digestive System in Aschelminthes.
Solution: **Ascaris** (roundworm), a member of Phylum Aschelminthes/Nematoda, is characterized by a distinctive, straight, alimentary canal with a highly developed, **muscular pharynx**. This muscular pharynx is specialized for the rhythmic pumping of food into the intestine.

Final Answer: Ascaris

138)

Explain Question: This question is asking which animal exhibits both polyp and medusa forms in its life cycle, a feature known as metagenesis.

Concept: This question is based on Phylum Coelenterata.

Solution: Only Obelia has well-defined life cycle involving both polyp and medusa, showing alternation of generations.

Final Answer: Option (2)

139) NCERT Pg. # 51

140) The correct answer is **1. Platyhelminthes.**

A. **Platyhelminthes (flatworms):** They have a single opening to their digestive system, called the gastrovascular cavity. This opening serves as both the mouth (for taking in food) and the anus (for expelling waste).

141)

Question Explanation: Identify the phylum with the water transport system.
Concept: Porifera
Solution: The **water transport** or**canal system** is a unique feature of **Porifera**.
Final Answer: 2

142)

Question Explanation: The question asks to identify the phylum to which 'Corals' belong.

Concept: Classification of Animal Kingdom (Phylum Cnidaria / Coelenterata)

Solution: Corals are marine invertebrates belonging to the phylum Cnidaria (class Anthozoa). They live in colonies of identical polyps and secrete a hard skeleton composed of calcium

carbonate that forms coral reefs.

Final answer: Option - 3 (Cnidaria)

143)

The space between the body wall and the alimentary canal lined by mesoderm is called the Coelom.

So, the correct answer is: 3. Coelom.

144) **Question Explanation:** The question asks to identify the two animals shown in the figures.

Concept: This question is based on common examples of Phylum Annelida.

Solution:
Figure (a) is a Nereis (a marine segmented worm) and Figure (b) is a Hirudinaria (leech). Both annelids.

The final Answer is option (3)

145) NCERT Pg. # 49

146) NCERT Pg. #44
Explanation: The chitinous exoskeleton is a rigid outer covering made of chitin (a polysaccharide) and proteins, providing protection, support, and muscle attachment.
It is found only in Arthropoda, making them the largest phylum in the animal kingdom.
The exoskeleton is periodically shed (molted) during growth in a process called ecdysis.
Correct Answer: Option 3. Arthropoda

147) NCERT XI Pg.# 53, Ist para

148) NCERT Pg. # 54 Para 4.2.9

149) NCERT Pg. # 54

150) NCERT (XIth) Pg. # 49

151)

Explain Question : Identify the phylum where nephridia perform excretion and osmoregulation.
Concept : This question is based on Annelida
Explanation : Nephridia are excretory and osmoregulatory organs found in segmented worms

(Annelida). Only Annelida possesses nephridia for both functions.
Final answer : OPTION 1 — Annelida

152) NCERT XI, Pg # 52

153)

NCERT XI Pg.# 41

154)

NCERT Pg. No. # 54

155)

Explain Question : Why Aschelminthes are called roundworms.

Concept : This question is based on Aschelminthes

Solution : Phylum Aschelminthes (Nematoda) includes worms with a cylindrical body. They are called roundworms due to their circular cross-section, unlike flatworms which are dorsoventrally flattened.

Final Answer : option (4) Both A and R are correct and R is the correct explanation of A

156) **Explain Question :** This question is asking about: Digestive system of Platyhelminthes.
Concept : Animal Kingdom → Phylum Platyhelminthes.
Solution : Platyhelminthes have an incomplete digestive system with a single opening that works as both mouth and anus. Thus, the assertion is true, and the reason correctly explains it.
Final answer : Option (1)

157)

Explain Question: Judge the correctness of two statements about Aschelminthes/Ascaris.
Concept: This question based on Aschelminthes
Solution: The correct answer is option 2.
Statement-I: True — Aschelminthes have a complete digestive tract with a well-developed muscular pharynx.
Statement-II: False — In Ascaris, females are longer (20–40 cm) than males (15–30 cm), and males have a curved posterior end.
Final Answer: 2

158)

Question Explanation: Reproduction and development in Balanoglossus
Concept: Hemichordata
Solution: Statement I is **correct**:**Balanoglossus** is **unisexual** (dioecious). Statement II is

incorrect : Fertilization is**external** and development is**indirect** (via tornaria larva).
Final Answer: 2

159) NCERT Pg. # 49

160) NCERT Pg. # 51

161)

Explain Question : This question asks about correct matching of given organisms with their examples.
Concept : This question is based on Animal Kingdom → Phylum Porifera & Coelenterata
Solution:

A. Sycon (Scypha) → Porifera → Match with ii
B. Physalia → Portuguese man of war → Match with iv
C. Gorgonia → Sea fan → Match with i
D. Adamsia → Sea anemone → Match with iii

Final answer : -3

162)

Explain Question:
Match echinoderm genera with their common names.
Concept: This question is based on Echinodermata
Solution:
• Echinus → Sea urchin (ii)
• Antedon → Sea lily (iii)
• Ophiura → Brittle star (i)
• Cucumaria → Sea cucumber (iv)
So the correct match is A-(ii), B-(iii), C-(i), D-(iv) → Option 2.
Final answer: 2. A-(ii), B-(iii), C-(i), D-(iv)

163)

Explain Question : Match each animal in Column I with its correct respiratory structure from Column II.
Concept : This question is based on Arthropoda, Annelida.
Explanation : The correct answer is option 3.
Prawn – Aquatic arthropod, respires through gills (v) for extracting oxygen from water.
Scorpion – Terrestrial arachnid, breathes through book lungs (iii), which are stacked lamellae for gas exchange.
Cockroach – Terrestrial insect, uses a network of tracheae (i) to deliver oxygen directly to tissues.
Earthworm – Annelid, respire cutaneously via moist cuticle (ii) since they lack specialized respiratory organs.
Thus, the correct matching is: A–v, B–iii, C–i, D–ii.

Final answer : 3

164) NCERT (XI) Pg. # 50,53,54

165) NCERT XIth Pg.#46,47

166) **Question Explanation:** Which group shows a blind sac type of body plan?
Concept: Blind Sac Body Plan
Solution

In a blind sac body plan, the digestive cavity has only one opening, which serves as both mouth and anus.

It is found in:

A. Coelenterata (Cnidaria)

B. Ctenophora

C. Platyhelminthes

Final Answer: Option (2)

167)

A. Annelida: This phylum is characterized by true metameric segmentation.

B. Arthropoda: Arthropods exhibit segmentation, but it is often more specialized, with segments grouped into distinct body regions like the head, thorax, and abdomen.

C. Chordata: Chordates show segmentation primarily in the mesoderm during embryonic development, forming structures which give rise to vertebrae, muscles, and nerves. In adults, this segmentation is evident in the vertebral column and musculature.

Therefore, segmentation is present in 3 of the listed phyla.

The correct option is 2.

168)

Explain Question : Identify the given special cell and its correct phylum.
Concept : This question is based on Cnidaria
Solution : The correct answer is **option 1.**

A. The figure shows a **Cnidoblast cell** (nematocyst), which is characteristic of **Coelenterata (Cnidaria)**. These cells help in anchorage, defense, and capturing prey.

Final Answer : 1

169)

Explain Question : Match phylum with corresponding examples and level of body organisation.
Concept : This question is based on levels of organisation – *Animal Kingdom*
Explanation :
A (roundworm) → Aschelminthes (ii) → Organ system level (b)
B (Hydra) → Coelenterata (iv) →Tissue level (a)
C (flatworm) → Platyhelminthes (i) → Organ level (c)
D(sponge) → Porifera (iii) → Cellular level (d)
So : A-ii-b, B-iv-a, C-i-c, D-iii-d
Final answer : 2

170)

Question Explanation: Identify the option that is completely correct without any exception.

Concept: Animal Classification

Solution: The correct answer is option 4

A. Option 1
Scypha (Porifera) does not show indirect development in the same way as Ctenophora, and fertilization pattern differs; not universally correct.

B. Option 2
Taenia lacks an alimentary canal, though flame cells are present for osmoregulation. So the statement is incorrect.

C. Option 3
Nereis has parapodia for locomotion, but Pheretima (earthworm) does not have parapodia; it moves using setae.

D. Option 4
Pinctada and Pila are molluscs; both have a mantle layer covering the body and possess gills for respiration. This statement is correct without exception.

Final Answer: Option 4

171) **Question Explanation :**
The question is asking how many of the given statements about sponges (phylum Porifera) are incorrect.

Concept :
phylum Porifera.

Solution :
(B) Incorrect – While most sponges are marine, not all are marine (some are freshwater). Also, sponges are mostly asymmetrical or radially symmetrical, not all are symmetrical.
(D) Incorrect – In sponges, fertilisation in internal and development is indirect (involves a Larvel stage).

Final Answer :
Option (3)

172)

Question Explanation: Identify which statements correctly describe phylum Coelenterata (Cnidaria).
Concept: Phylum – Coelenterata
Solution: The correct answer is option 3.

A. (A) Aquatic, mostly marine, sessile or free-swimming, radially symmetrical - True for cnidarians like Hydra, Aurelia, Obelia.

B. (B) Central gastrovascular cavity with a single opening (mouth) on hypostome - Correct characteristics.

C. (C) Digestion is both extracellular (in GVC) and intracellular - True for Cnidaria.

D. (D) Examples: Sycon, Spongilla, Euspongia - These are Porifera, not Coelenterata.

So A, B, C are correct; D is incorrect.

Final answer: Option (3)

173)

Question Explanation: Incorrect statement for Balanoglossus.
Concept: Hemichordata
Solution: The incorrect statement is that **flame cells are excretory organs**. The excretory organ is the**proboscis gland**.
Final Answer: 1

174)

Question explanation: Identify the statement(s) that are **not characteristic** of phylum Echinodermata.
Concept: Animal Kingdom – Echinodermata

Solution:

A. **(i) Water vascular system** → Correct (present in echinoderms).

B. **(ii) Protostomic origin** → Incorrect (echinoderms are **deuterostomes**, not protostomes).

C. **(iii) Endoskeleton** → Correct (calcareous endoskeleton present).

D. **(iv) Open circulatory system** → Considered correct in echinoderms (circulatory system is reduced/open).

Incorrect statement: Only (ii)

Final answer: Option 3

175) NCERT (XI) Pg. # 51,52 (H)

176) NCERT XI, Page No. # 51 IInd Paragraph

177)

Question Explanation: Identify the organism that is not a sponge.
Concept: Porifera, Cnidaria
Solution: Sea anemone (**Adamsia**) is a **Coelenterate** , not a sponge.
Final Answer: 4

178) NCERT Page No. # 53, 54

179)

Question explanation
The question is asking how many of the listed animals belong to the phylum Echinodermata.

Concept
Echinodermata.

Solution
Animals in Echinodermata are Asterias, Echinus, Sea lily, Ophiura, and Cucumaria. The others belong to different phyla (Saccoglossus – Hemichordata, Cuttle fish – Mollusca, Sea hare – Mollusca, Culex – Arthropoda, Chiton – Mollusca). Total = 5.

Final answer
Option 1

180) **Question Explanation**: Identify the correct labeling of germ layers and coelom in a sectional view of a triploblastic coelomate animal.

Concept: The question is based on coelom.

Solution: The correct answer is option 3.

A. Part-A: Ectoderm (outermost layer)
B. Part-B: Mesoderm (middle layer)
C. Part-C: Coelom (body cavity between mesodermal layers)
D. Part-D: Endoderm (inner lining of body wall).

This structure is typical of triploblastic coelomates, such as annelids, molluscs, and chordates.

Final answer: Option 3